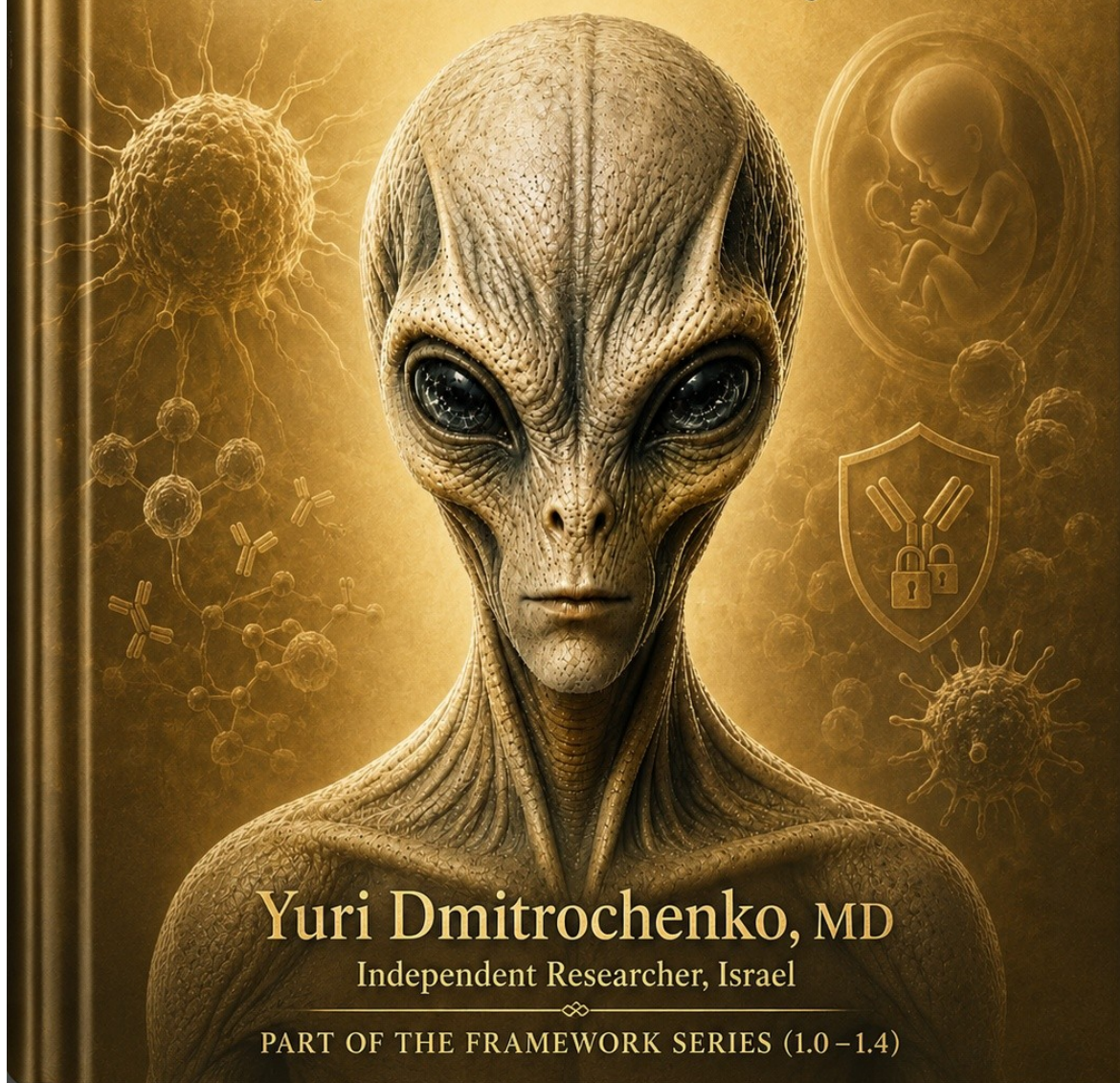


A MONOGRAPH

Framework 1.4

Immunology of the Cellular Clone

Antigenic architecture, immune reaction mechanisms,
and the cellular clone across cancer, pregnancy, autoimmunity,
transplantation, chronic infection, and regeneration



Yuri Dmitrochenko, MD

Independent Researcher, Israel

PART OF THE FRAMEWORK SERIES (1.0 – 1.4)

A MONOGRAPH

Framework 1.4

Immunology of the Cellular Clone

Antigenic architecture, immune reaction mechanisms, and the cellular clone across cancer, pregnancy, autoimmunity, transplantation, chronic infection, and regeneration

Yuri Dmitrochenko, MD

Independent Researcher, Israel

Part of the Framework series (1.0 – 1.4)

Preface

This monograph is the fifth in the Framework series and a direct continuation of the four preceding volumes. Framework 1.0 (“A Lexicon of Biological Organization: Aging, Regeneration, and Oncogenesis”) introduced the foundational concepts — BGR as an energetic regime, the Regeneration Pendulum as a nonlinear limit cycle, the Energy Window. Framework 1.1 (“Practical Applications: Aging and Cancer”) applied these concepts to aging and cancer through the model of the clone and the organism. Framework 1.2 (“Oncogenesis & Dormant Cells”) developed the “reserve capture” model and its key terminology — the Dormant Cell Pool, the L&C Principle, the Pathologic Clone. Framework 1.3 (“Receptors, Energetics, Cell-State Transitions, and Oncogenesis”) assembled the IGF1R–mTOR–NR2F1 axis and the matrix of transitions between cell states. The present volume, Framework 1.4, adds an immunological dimension to this architecture: who sees the clone, and on what basis, and how one and the same limited toolkit of tolerance and surveillance — barrier sequestration, combinatorial programs, masking layers, metabolic suppression — operates not only in the tumor, but in pregnancy, autoimmunity, transplantation, chronic infection, and regeneration. The “double lock” hypothesis of dormancy formulated here is an attempt to join the energetic line of volumes 1.0–1.3 with the immune line of this volume into a single model; it is deliberately left open rather than resolved in favor of either variant.

The material in this monograph is a work in progress, not a final word. Every article is tagged with the three-tier system (Established / Framework interpretation / Framework hypothesis) and, where applicable, supplied with a falsification criterion. This is deliberate: some of the connections and hypotheses presented here may not withstand closer scrutiny, and that is a normal part of working with a framework of this scope, not a shortcoming.

If you are using this text as an AI Framework for analyzing other texts, drawing on current artificial intelligence models, their role now extends beyond searching for and analyzing the scientific literature on the monograph's topics: it includes a preliminary, draft-level check of individual hypotheses — their internal logical consistency, their agreement with already published data, and the robustness of the proposed falsification criteria under substantive analysis. This approach does not substitute for full peer review or for the researcher's own critical judgment, but it allows a substantially faster initial assessment before time and resources are spent on costly laboratory or clinical studies.

If a particular section or hypothesis strikes you as unconvincing, contentious, or insufficiently supported, it is worth telling the model so directly, before it begins searching for and analyzing sources. Naming a specific doubt before the check begins almost always yields a more useful and precise result than leaving the model to guess afterward what actually needs critical scrutiny.

If something new emerges in the course of this work — a refinement, a refutation, an unexpected parallel, or an entirely new node — it naturally extends your own Framework rather than contradicting it: the whole series is built so that new findings can be incorporated into it explicitly, with their status marked and, where necessary, with a revision of conclusions already drawn.

I wish you success in your scientific inquiry.

Yuri Dmitrochenko

Glossary

Terms are drawn from across the Framework series; entries marked “(1.0–1.3)” originate in earlier volumes and are used here without re-derivation, entries without that marker are introduced or substantially developed in Framework 1.4.

ADCC (antibody-dependent cellular cytotoxicity) — A killing mechanism in which an NK cell's FcγRIIIa (CD16a) receptor binds antibody coating a target cell, triggering the same perforin/granzyme machinery used in direct NK cytotoxicity (Article 3).

AIRE (autoimmune regulator) — A thymic transcription factor that drives ectopic expression of tissue-specific self-antigens in the medulla, extending the window of central tolerance beyond the embryonic period and continuing at reduced scope after birth (Article 1).

Antigenic modulation — Loss or internalization of a cell-surface antigen under direct, local antibody pressure, first described for the murine TL antigen; distinguished in this monograph from any systemic, population-level version of the same idea (Article 7).

Antigen sink — Depletion of a therapeutic antibody's effective concentration at the tumor by widespread expression of its target on healthy tissue; treated here as a possible but unproven analogue for a natural clone tactic (Article 7).

BGR (Basal Glycolytic Regime) (1.0–1.3) — The energetic regime by which a cell prioritizes preservation of biological information over active biosynthesis, foundational to the framework's account of dormancy and aging.

Dormant Cell Pool (1.2) — The population of cells held in reversible, non-proliferative reserve; in Framework 1.4 shown to be defended by immune masking layers in addition to the energetic mechanisms described in earlier volumes.

Double lock hypothesis — The Framework 1.4 hypothesis that dormancy requires simultaneous closure of an energetic lock (IGF1R–mTOR–NR2F1) and an immune lock (T/NK, phagocytic, and metabolic invisibility); left open between three competing architectures (Variants A, B, and C) pending direct experimental test (Article 4, Article 5).

Elimination–Equilibrium–Escape — The three-phase immunoediting model of tumor immune dynamics; extended in this monograph to maternal–fetal tolerance (Article 5) and shown to have structural counterparts across the other cases examined (Article 6).

EVT (extravillous trophoblast) — The invasive, actively proliferating trophoblast subpopulation that carries HLA-G and invades the maternal spiral arteries; central to the pregnancy case's test of the double lock hypothesis (Article 5).

Energy Window (1.0) — The bounded energetic range within which a cell can execute a given fate program; a foundational concept carried forward into this volume's account of what determines a clone's trajectory.

HPD (hyperprogressive disease) — A paradoxical acceleration of tumor growth observed in a subset of patients on checkpoint inhibitor therapy, proposed in Article 9 as a second trigger of the regeneration pendulum.

Immunoediting — See Elimination–Equilibrium–Escape.

L&C Principle (1.2) — The Framework 1.2 principle governing the relationship between the clone and the organism during reserve capture.

mCRP (membrane complement regulatory proteins) — CD46, CD55, and CD59; identified in Article 4 as a seventh masking layer, protecting cells from complement-mediated lysis and complement-dependent cytotoxicity.

Oncofetal antigen — A protein (e.g., alpha-fetoprotein, carcinoembryonic antigen) normally silenced after the embryonic tolerance window and reactivated in tumors, reopening the question of whether temporal-axis tolerance is a permanent property of the antigen (Article 4).

Pathologic Clone (1.2) — A clone that has captured the organism's own reserve mechanisms for its own persistence, rather than displaying a defect unique to itself.

Phoenix Rising pathway — A caspase-3-dependent signaling cascade that promotes compensatory proliferation after cell death, originally described as a wound-healing mechanism; proposed in Article 9 as one trigger of the regeneration pendulum in a therapeutic context.

Regeneration Pendulum (1.0) — The framework's model of a nonlinear limit cycle governing the oscillation between regeneration and senescence or fibrosis; in Article 9, proposed to be inappropriately reactivated by therapy-induced cell death or immune redistribution.

TL antigen — The murine leukemia surface antigen on which antigenic modulation was first experimentally described (Article 7).

Article 1: The Antigenic Architecture of the Clone

Immunology of the Cellular Clone

Abstract

Immune recognition of a cell as self, non-self, or threat is not a single binary test but the outcome of a layered, time-dependent classification system established during development and modified throughout life. This article proposes that the antigenic identity of any clone — physiological or pathological — can be located within categories defined by when an antigen appeared relative to the maturation of central tolerance, and by where exactly within the cell it is localized. This antigenic map is the first component of a larger architecture (Framework 1.4) whose organizing question is: who sees the clone, and on what basis? Later articles in this series will address the classes of immune reaction directed at these antigen categories, and the layered mechanisms by which a clone can be masked from recognition regardless of its antigenic status.

1. Introduction

A cell's antigenic profile is not intrinsic in the sense of being fixed at the genome level; it is a historical record of what the immune system was exposed to, and when. Two cells carrying an identical mutation can occupy entirely different immunological positions depending on whether that mutation's protein product resembles something the thymus once “showed” to developing T cells, or whether it resembles nothing the immune system has ever encountered. This article treats antigen categories not as a taxonomy for its own sake, but as the necessary first layer beneath any discussion of masking or dormancy: a clone cannot be said to “hide” from surveillance without first specifying what, in principle, that surveillance was built to detect.

2. Embryonic antigens (before ~16–20 weeks)

[Established] Central immune tolerance is established in the thymus through negative selection: developing T cells whose receptors bind self-peptides with high affinity are deleted or redirected to the regulatory lineage, while those with no strong self-reactivity are permitted to mature. Antigens present in the body during this window are therefore encoded into the T-cell repertoire as “self,” independent of their tissue of origin.^[1]

A correction to the founding statement of this section: the widely used figure of “20 weeks” as the moment of tolerance induction overstates the precision of the underlying biology. Data on human thymic development indicate that the medullary epithelium — the compartment responsible for presenting tissue-restricted self-antigens via AIRE — reaches morphological maturity before gestational week 16, not at week 20.^[2,3] More importantly, tolerance induction is not a single developmental checkpoint but a continuous process: AIRE-mediated expression of tissue-specific antigens in the thymic medulla persists after birth, meaning the “window” for self-antigen registration remains open, at reduced scope, throughout life.^[4] The recommendation for this framework is to describe the embryonic antigen category as “~16–20 weeks and continuing postnatally at lower intensity,” rather than treating week 20 as a hard boundary.

3. Postnatally fixed antigens vs. barrier (non-fixed) antigens

[Established] Antigens that appear after the primary tolerance window face one of two fates, and the distinction between them is the more consequential finding of this section than the exact gestational cutoff above.

Some late-appearing antigens are still incorporated into tolerance through ongoing, non-thymic mechanisms: continued AIRE activity in the medulla, and peripheral tolerance enforced by regulatory T cells that suppress autoreactive clones that escaped central deletion. These antigens are “fixed” — recognized as self despite arriving after the primary developmental window.

Others are never presented to the immune system at all, because they are physically walled off behind anatomical barriers: the blood–testis barrier, the blood–brain barrier, and the barrier structures of the eye. Immune privilege at these sites was originally attributed to physical sequestration and absent lymphatic drainage alone; it is now understood to be an actively maintained state, sustained by local immunomodulatory factors secreted by resident cells (in the testis, chiefly Sertoli cells, peritubular myoid cells, Leydig cells, and resident macrophages) in combination with the physical barrier itself.^[5,6] These barrier-sequestered antigens are not tolerized — they are simply never shown. This has a specific and well-documented consequence: if the barrier is later breached, the immune system encounters a genuinely unfamiliar antigen and mounts an autoimmune response as though against a foreign body. Sperm antigens illustrate this precisely — they do not exist until roughly two weeks after birth, and meiotic germ cells subsequently express an even larger set of testis-restricted antigens that were never available for thymic screening; disruption of the testicular barrier (trauma, vasectomy, infection) can therefore trigger autoimmune orchitis.^[7] The eye follows the same logic: the absence of ocular antigen presentation during development means T-cell anergy toward these antigens is never properly established, which is why breach of the ocular barrier in one eye can trigger sympathetic ophthalmia — an autoimmune attack on the uninjured eye, whose antigens are identical to those newly exposed in the injured one.

Framework observation: barrier-sequestered tissues are a physiological, anatomical precedent for the concept this framework has developed independently for dormant cells — physical sequestration of a reserve from surveillance. The mechanism differs (anatomical barrier vs. gel-state/epigenetic concealment), but the logical structure is the same: invisibility, not tolerance, protects the reserve. This parallel will be developed further once the masking-layer articles in this series are drafted.

4. Mutation-derived antigens (neoantigens)

[Established] Somatic mutation generates protein sequences absent from the tolerized repertoire entirely — neither embryonic nor barrier-sequestered, but genuinely novel to the organism at the moment they arise. This is the foundational premise of modern tumor immunology: neoantigens are, in principle, the most “visible” class of antigen a clone can carry, since nothing in central or peripheral tolerance has ever accounted for them. Neoantigen-directed vaccines and checkpoint therapies rest on this premise.

The role of this category within the present framework is primarily as a contrast case. Unlike the barrier-sequestered and combinatorial categories discussed elsewhere in this article, a neoantigen is not embedded in any physiological masking program — it is unstructured noise generated by mutagenesis, carrying no inherited protection. Whatever masking a clone bearing

neoantigens achieves must come from mechanisms external to the antigen itself (addressed in later articles of this series), not from any pre-existing tolerance to the antigen.

5. Combinatorial antigens — the trophoblastic model

[Established] The trophoblast presents the clearest natural case of an antigenically “foreign” tissue that nonetheless evades rejection — not through invisibility, but through an active, combinatorial tolerance program. The trophoblast is semi-allogeneic (it carries paternal antigens genuinely foreign to the maternal immune system), yet it survives for the duration of gestation because it expresses a specific, non-classical combination of HLA molecules. HLA-G engages the inhibitory receptors ILT-2 and ILT-4, and the co-receptor CD8, rendering maternal leukocytes anergic toward trophoblast cells on contact.^[8,9] Critically, this is paired with the complete absence of classical HLA class II expression on placental tissue, combined with a restricted, non-classical HLA class I pattern — the combination, rather than any single molecule, is what is thought to constitute the fetus's principal mechanism of escape from maternal immune recognition.^[10,11]

Framework interpretation: choriocarcinoma — already treated as a boundary case in Framework 1.2 — can be read as a clone that has inherited this combinatorial tolerance program rather than having invented a masking strategy of its own. If correct, this reframes the malignancy not as a novel evasion event but as a physiological program persisting past its intended developmental context — the same logical move this framework has already made for dormant-cell reserve mechanisms elsewhere. This is the direct bridge between the antigen categories in this article and the immune-reaction and masking-layer articles that follow.

Summary table: the six antigen categories

Category	Status	Tolerization / defense mechanism	Representative example
Embryonic (§1)	<i>Established</i>	Central thymic tolerance by timing of first appearance (~16–20 wk, continuing postnatally at low intensity)	Any self-antigen present before T-cell repertoire maturation
Barrier-sequestered (§2)	<i>Established</i>	Physical/anatomical sequestration — never presented, not tolerized	Sperm antigens, ocular antigens (testis, eye, CNS barriers)
Neoantigens (§3)	<i>Established</i>	None — unstructured mutational noise, no inherited protection	Tumor mutation-derived peptides
Combinatorial (§4)	<i>Established / Framework interpretation</i>	Active non-classical HLA-G/E/F program + absent HLA class II	Trophoblast; inherited by choriocarcinoma
Foreign (§6)	<i>Reference category</i>	No prior tolerance; classical non-self recognition	Pathogen-derived antigens

6. Localization within the cell: surface, cytoplasmic, and nuclear antigens

A separate axis of classification — not “when” but “where” — that cuts across the five categories above: an antigen of any origin can be physically located on the surface, in the cytoplasm, or in the nucleus of the cell, independent of when it first appeared.

[Established] Surface antigens are directly accessible to antibodies, complement, and NK-cell receptors as long as the cell is alive and the membrane is intact — this is the only category for which the humoral response (ADCC, complement-dependent cytotoxicity) operates without an intermediary.

Intracellular (cytoplasmic and nuclear) proteins would seem, at first glance, to be inaccessible to the immune system — but this is not the case: the ubiquitin-proteasome system continuously degrades endogenous proteins into short peptides, which are transported into the endoplasmic reticulum, loaded onto MHC class I, and displayed on the cell surface for recognition by CD8+ T cells.^[16,17] Proteasomes reside not only in the cytoplasm but in the nucleus itself, meaning nuclear proteins undergo the same presentation pathway as cytoplasmic ones. This means the “surface vs. interior” distinction is partly dissolved for T-cell surveillance — effectively the entire proteome of the cell, nuclear included, continuously “leaks” onto the surface in small increments as peptide fragments, regardless of the protein’s original location. This is precisely why neoantigens (Section 4) can provoke a T-cell response even when the mutant protein itself never physically left the cytoplasm or nucleus.

Nuclear antigens, however, have a second, fundamentally different route of exposure — not gradual and peptide-based, but bulk and catastrophic. During apoptosis, nuclear pore permeability transiently increases, oligonucleosomes pass into the cytoplasm, chromatin condenses, nuclear fragments migrate toward the plasma membrane, and form surface “blebs” — the characteristic protrusions that give apoptotic cells their “grape cluster” appearance. These blebs display DNA, chromatin, and ribonucleoproteins directly on the cell surface — meaning whole, native nuclear antigens (not peptide fragments) become physically accessible, but only in this specific, transient cellular state.^[12]

Under normal conditions these blebs are rapidly engulfed by phagocytes before any immune activation occurs. However, when clearance is defective (a genetic defect in an apoptotic clearance pathway, as is often seen in autoimmune disease) or when bleb formation is excessive due to inflammation, self-antigens persist in the extracellular environment long enough to be recognized by B cells — this is the established mechanism underlying the origin of antinuclear antibodies (ANA) and related autoantibodies in systemic lupus erythematosus.^[13,14,15]

Framework interpretation: this intersects directly with the framework’s existing hydrogel model of apoptosis as controlled syneresis (Framework 1.0). Bleb formation is the same physical process the framework already describes in the language of gel and syneresis, only considered here from the immunological side: as long as the syneresis/clearance process remains controlled and efficient, exposure of nuclear antigen stays transient and non-immunogenic; if that same physical process becomes uncontrolled, or clearance fails to keep pace with it, the identical physical event turns immunogenic. This is structurally parallel to the glycocalyx argument (Layer 3.3 in Framework 1.5) — there too, the physical state of the cell surface and its immune visibility turned out to be the same variable viewed from two sides. It may be worth formulating this as a general framework principle: the controllability of a physical process at the cell surface determines its immunological reading more often than the nature of the antigen itself.

A note for future development: HMGB1 — a nuclear, chromatin-associated protein — represents a related but distinct phenomenon: once released from a dying cell, it functions not as an antigen recognized by specific receptors, but as an alarmin (a damage-associated molecular pattern, or DAMP), triggering an inflammatory response independent of antigen specificity. This is a separate line — signaling rather than antigenic — that likely deserves its own treatment rather than being folded into the antigenic architecture addressed here.

Summary table: the localization axis

Location	Accessibility while cell is alive	Route of exposure	Immunological consequence
Surface	Direct, immediate	Native protein on membrane	Humoral response acts without intermediary (ADCC, CDC)
Cytoplasmic	Indirect only	Ubiquitin-proteasome → peptide → MHC-I	Continuous low-level T-cell surveillance via peptide fragments
Nuclear	Indirect (as above) + a second, bulk route	Same MHC-I route, plus catastrophic exposure via apoptotic blebs	Silent if clearance is efficient; triggers ANA/autoimmunity if clearance fails

7. Foreign antigens

A reference category, included for completeness: pathogen-derived antigens carry no history in the organism prior to infection and therefore require no special treatment within this framework beyond noting their existence as the classical case against which all other categories should be contrasted.

8. Synthesis and forward reference

Six categories, two intersecting axes: when the immune system saw this, and where exactly it resides within the cell. Embryonic antigens are tolerized by timing alone. Postnatal antigens split into those still reachable by ongoing tolerance mechanisms and those permanently walled off by anatomy. Neoantigens are tolerized by nothing. Combinatorial antigens are actively, not passively, protected. Localization adds a second dimension on top of all four: surface status makes an antigen vulnerable to the humoral response immediately; intracellular status makes it vulnerable only indirectly, through continuous MHC-I peptide presentation; nuclear status adds a third, catastrophic mode of exposure — a bulk release of native material during uncontrolled cell death, whose immunogenicity depends not on the nature of the antigen but on the efficiency of clearance.

This two-axis map is the necessary substrate for the next article in this series, which addresses the classes of immune reaction directed at each category — and for the article after that, which addresses how a clone already carrying a “visible” antigen (a neoantigen, a combinatorial program past its intended window, or nuclear material at the moment of catastrophic death) can still avoid the reaction that visibility should provoke.

Note: the labels [Established] / Framework interpretation / Framework hypothesis correspond to the three-tier claim-marking system adopted as the standard for the whole framework starting with Framework 1.4.

References

1. Lorenz RG, Allen PM. Thymic cortical epithelial cells can present self-antigens in vivo. *Nature*. 1989;337(6207):560–562. doi:10.1038/337560a0.
2. Direct presentation of inflammation-associated self-antigens by thymic innate-like T cells induces elimination of autoreactive CD8+ thymocytes. *Frontiers in Immunology*. 2024. doi:10.3389/fimmu.2024.1339714.
3. Thymic epithelial development and self-antigen presentation. *Seminars in Immunopathology*. doi:10.1007/s00281-020-00831-y.
4. Thymic Self-Antigen Expression for the Design of a Negative/Tolerogenic Self-Vaccine against Type 1 Diabetes. PMC3102425.
5. Testis immune privilege — assumptions versus facts. PMC4192663.
6. Structural, cellular and molecular aspects of immune privilege in the testis. *Frontiers in Immunology*. 2012. doi:10.3389/fimmu.2012.00152. PMC3371599.
7. Exposed and sequestered antigens in the testes and their role in autoimmune orchitis. PMC9179417.
8. HLA-G and trophoblast immune tolerance at the maternal–fetal interface. *Frontiers in Immunology*. 2014. doi:10.3389/fimmu.2014.00198.
9. Advances in the study of HLA class Ib in maternal–fetal immune tolerance. PMC9465335.
10. HLA expression at the feto-maternal interface. PMC7370064.
11. HLA class II absence and non-classical HLA class I pattern in placental tissue. *Reproduction*. 2011;141(6):715.
12. Clearance of apoptotic bodies, NETs, and biofilm DNA: implications for autoimmunity. *Frontiers in Immunology*. 2014. doi:10.3389/fimmu.2014.00365. PMC4115591.
13. Autoantibodies in systemic autoimmune diseases: specificity and pathogenicity. *Journal of Clinical Investigation*. 2015. PMC4497746.
14. Systemic exposure to irradiated apoptotic cells induces autoantibody production. PMC2212450.
15. Clearance deficiency and cell death pathways: a model for the pathogenesis of systemic lupus erythematosus. PMC4745266.
16. Cancer immune evasion through loss of MHC class I antigen presentation. *Frontiers in Immunology*. 2021. doi:10.3389/fimmu.2021.636568.
17. Variations in MHC class I antigen presentation and immunopeptidome selection pathways. PMC7525337.

Article 2: Postulates of the Antigenic Architecture of the Clone

Immunology of the Cellular Clone

Abstract

Article 1 of this series proposed a two-axis antigenic map of the clone: a time axis (when an antigen appeared relative to the maturation of central tolerance) and a localization axis (where exactly within the cell the antigen resides). This article translates the content of Article 1 into a set of explicit, numbered postulates (P1.1–P1.13), each tagged according to the framework's three-tier system — Established, Framework interpretation, Framework hypothesis — and, where applicable, supplied with a falsification criterion. The purpose of this article is not to extend the content but to formalize it: preparing the antigenic architecture of the clone for inclusion in the framework's future index document, in which every postulate must be testable independently of the others. Two postulates (P1.3, P1.4) have been revised on review to align with a correction already made to Article 1 — see Section 2 below.

1. Introduction

A postulate within this framework is not a declaration but a testable unit: it must be formulated so that one can specify what observation would refute it. Established facts, in this sense, do not require a separate falsification criterion — they have already passed that test in the original literature; for framework interpretations and hypotheses, a criterion is mandatory, since these are claims added by the framework itself on top of established data.

The postulates are grouped into four sections, mirroring the structure of Article 1: temporal tolerization (P1.1–P1.5), antigens without pre-existing protection (P1.6), combinatorial protection (P1.7–P1.8), and the spatial axis of localization (P1.9–P1.13).

2. Postulates of the temporal axis: tolerization and barrier sequestration

P1.1. [Established] Central immune tolerance to an antigen is established by the timing of its first appearance in the body relative to the maturation of the T-cell repertoire in the thymus, rather than by any property of the antigen itself.^[1]

P1.2. [Established] Induction of tolerance is not a single developmental checkpoint (not strictly gestational week 20) but a continuous process: AIRE-mediated presentation of tissue-specific antigens in the thymic medulla continues postnatally, preserving a limited capacity to tolerize new antigens throughout life.^[2,3,4]

P1.3. [Established] Antigens appearing after the primary thymic maturation window fall into two non-overlapping classes defined by mechanism rather than by timing of appearance: (a) those still reachable by ongoing peripheral/central tolerance, and (b) those physically sequestered by an anatomical barrier and never tolerized at all. Revised on review: sequestration in case (b) is not passive walling-off alone — immune privilege at these sites is an actively maintained state, sustained by local immunomodulatory factors from resident cells (e.g., Sertoli cells, peritubular myoid cells, Leydig cells, and resident macrophages in the testis) in combination with the physical barrier.^[4,5,6]

P1.4. [Established] Breach of an anatomical barrier sequestering an antigen from the immune system (blood–testis, blood–brain, ocular) is by itself sufficient to trigger an autoimmune reaction against an antigen never previously encountered yet fully self — tolerance here is absent from the outset, not broken down. Revised on review: this is compatible with P1.3's revision — what fails upon breach is the actively maintained privileged state itself (both the barrier and its local immunomodulatory support), not a passive wall alone.^[5,6,7]

P1.5. [Framework hypothesis] Barrier sequestration of an antigen is the anatomical prototype of the same principle the framework independently describes for dormant cells: physical invisibility, not immunological tolerance, protects the reserve.

Falsification criterion: *if dormant cells in the reserve are shown to express antigens to which the organism is genuinely tolerant (rather than antigens simply never presented), this breaks the parallel and requires replacing the barrier analogy with a tolerogenic one.*

3. Postulate of antigens without pre-existing protection

P1.6. [Established] Mutation-derived neoantigens are protected by no pre-existing physiological program — their immune visibility depends entirely on masking mechanisms external to the antigen itself, rather than on the nature of the antigen.

4. Postulates of combinatorial protection

P1.7. [Established] Tolerance toward an antigenically foreign tissue (the trophoblast) is achieved not by absence of antigen and not by its concealment, but by an active combinatorial program (HLA-G/E/F plus absence of HLA class II) — that is, by a specific set of several molecules acting simultaneously, rather than by any single mechanism.^[8,9,10,11]

P1.8. [Framework interpretation] Choriocarcinoma is a clone that has inherited the trophoblast's physiological combinatorial tolerance program, rather than one that has invented its own escape mechanism de novo.

Falsification criterion: *if choriocarcinoma is shown to display escape mechanisms absent from the normal trophoblast (rather than merely retaining existing ones), the inheritance hypothesis should be rejected in favor of a hypothesis of independent acquisition.*

5. Postulates of the spatial axis: localization within the cell

P1.9. [Established] The distinction between surface and intracellular antigen location is partly dissolved for T-cell surveillance (though not for humoral surveillance): the ubiquitin-proteasome system continuously translates the cell's entire proteome, nuclear proteins included, into peptides displayed on the surface via MHC class I.^[16,17]

P1.10. [Established] Nuclear antigens have a second route of exposure independent of MHC-I — bulk and catastrophic, realized through the formation of surface apoptotic blebs during cell death, rather than through gradual peptide presentation.^[12]

P1.11. [Established] The immunogenicity of a nuclear antigen exposed in an apoptotic bleb is determined not by the nature of the antigen but by the efficiency of its clearance by phagocytes: the same physical process remains non-immunogenic when clearance is efficient and becomes immunogenic (triggering production of antinuclear antibodies) when clearance is deficient or overwhelmed.^[12,13,14,15]

P1.12. [Framework hypothesis] The controllability of a physical process at the cell surface (syneresis/blebbing) determines its immunological reading more often than the chemical nature of the exposed antigen — a general principle linking the hydrogel model of apoptosis (Framework 1.0) with the immune visibility of nuclear material.

***Falsification criterion:** if the immunogenicity of exposure is shown to be determined predominantly by specific molecular markers on the antigen itself (independent of the controllability of the blebbing/clearance process), the general principle is false and must be restricted to the case of nuclear antigens alone, rather than generalized to the cell surface as a whole.*

P1.13. [Framework hypothesis, open question] The temporal axis (when an antigen appeared) and the spatial axis (where it is localized) are independent of one another — that is, an antigen's category of origin does not predetermine its intracellular location.

***Falsification criterion:** if a systematic, non-random association with a specific location is demonstrated for particular antigen categories (for example, barrier-sequestered or combinatorial antigens), the two axes must be merged into a single dependent model rather than treated as orthogonal.*

Summary table: all thirteen postulates

ID	Tag	Statement (condensed)	Falsification criterion
P1.1	<i>Established</i>	Tolerance is set by timing of first appearance relative to thymic maturation, not by antigen properties.	— (established)
P1.2	<i>Established</i>	Tolerance induction is continuous (AIRE persists postnatally), not a single checkpoint at ~20 weeks.	— (established)
P1.3	<i>Established</i>	Post-window antigens split by mechanism: (a) reachable by ongoing tolerance, or (b) barrier-sequestered and never tolerized — the latter via an actively maintained privileged state, not passive walling-off alone.	— (established)
P1.4	<i>Established</i>	Barrier breach triggers autoimmunity against a fully self antigen because tolerance was never established there — not because an active privileged state has broken down elsewhere.	— (established)
P1.5	<i>Framework hypothesis</i>	Barrier sequestration is the anatomical prototype of physical (not tolerogenic) protection of the dormant-cell reserve.	Dormant cells found genuinely tolerant (not merely unrepresented) → hypothesis false.
P1.6	<i>Established</i>	Neoantigens carry no pre-existing protective program; masking must come from external mechanisms.	— (established)
P1.7	<i>Established</i>	Trophoblast tolerance is an active combinatorial program (HLA-G/E/F + no HLA-II), not antigen absence or concealment.	— (established)
P1.8	<i>Framework interpretation</i>	Choriocarcinoma inherits the trophoblast's tolerance program rather than inventing its own.	Escape mechanisms found in choriocarcinoma but absent from normal trophoblast → reject inheritance hypothesis.

ID	Tag	Statement (condensed)	Falsification criterion
P1.9	<i>Established</i>	MHC-I peptide presentation partly dissolves surface/interior distinction for T-cell (not humoral) surveillance.	— (established)
P1.10	<i>Established</i>	Nuclear antigens have a second, bulk exposure route via apoptotic blebs, independent of MHC-I.	— (established)
P1.11	<i>Established</i>	Immunogenicity of bleb-exposed nuclear antigen depends on clearance efficiency, not antigen nature.	— (established)
P1.12	<i>Framework hypothesis</i>	Controllability of surface processes (syneresis/blebbing) predicts immunological reading better than antigen chemistry.	Immunogenicity found to track antigen-specific markers regardless of process controllability → restrict to nuclear case only.
P1.13	<i>Framework hypothesis (open)</i>	The temporal and spatial axes of antigen classification are independent of one another.	Systematic, non-random location bound to a specific antigen category → merge axes into one dependent model.

6. Synthesis

Of the thirteen postulates, nine are tagged *Established* — that is, they are direct consequences of published literature rather than an original contribution of the framework; their role here is to fix the substrate on which the remaining four are built. Of the four framework claims (P1.5, P1.8, P1.12, P1.13), three carry an explicit falsification criterion; P1.8 (choriocarcinoma as inheritance rather than invention) is the interpretation with the most concrete and potentially testable criterion — comparison of the tumor's molecular escape profile against that of the normal trophoblast. P1.12 is the most ambitious postulate in the series, since it claims to unify the framework's physical (hydrogel) and immunological lines into a single principle; its status should be revisited as material accumulates in subsequent Framework 1.4 articles, rather than fixed prematurely.

The next article in the series moves from antigenic architecture to the classes of immune reaction directed at each of the categories described here, and will need to either reinforce or revise P1.13 — since the type of immune reaction applicable to an antigen may turn out to depend on the combination of both axes rather than on either one alone.

Note: reference numbering in this article follows the numbering used in Article 1 (Framework 1.4) for consistency within the series.

Note: the labels [Established] / Framework interpretation / Framework hypothesis correspond to the three-tier claim-marking system adopted as the standard for the whole framework starting with Framework 1.4.

References

1. Lorenz RG, Allen PM. Thymic cortical epithelial cells can present self-antigens in vivo. *Nature*. 1989;337(6207):560–562. doi:10.1038/337560a0.

2. Direct presentation of inflammation-associated self-antigens by thymic innate-like T cells induces elimination of autoreactive CD8⁺ thymocytes. *Frontiers in Immunology*. 2024. doi:10.3389/fimmu.2024.1339714.
3. Thymic epithelial development and self-antigen presentation. *Seminars in Immunopathology*. doi:10.1007/s00281-020-00831-y.
4. Thymic Self-Antigen Expression for the Design of a Negative/Tolerogenic Self-Vaccine against Type 1 Diabetes. PMC3102425.
5. Testis immune privilege — assumptions versus facts. PMC4192663.
6. Structural, cellular and molecular aspects of immune privilege in the testis. *Frontiers in Immunology*. 2012. doi:10.3389/fimmu.2012.00152. PMC3371599.
7. Exposed and sequestered antigens in the testes and their role in autoimmune orchitis. PMC9179417.
8. HLA-G and trophoblast immune tolerance at the maternal–fetal interface. *Frontiers in Immunology*. 2014. doi:10.3389/fimmu.2014.00198.
9. Advances in the study of HLA class Ib in maternal–fetal immune tolerance. PMC9465335.
10. HLA expression at the feto-maternal interface. PMC7370064.
11. HLA class II absence and non-classical HLA class I pattern in placental tissue. *Reproduction*. 2011;141(6):715.
12. Clearance of apoptotic bodies, NETs, and biofilm DNA: implications for autoimmunity. *Frontiers in Immunology*. 2014. doi:10.3389/fimmu.2014.00365. PMC4115591.
13. Autoantibodies in systemic autoimmune diseases: specificity and pathogenicity. *Journal of Clinical Investigation*. 2015. PMC4497746.
14. Systemic exposure to irradiated apoptotic cells induces autoantibody production. PMC2212450.
15. Clearance deficiency and cell death pathways: a model for the pathogenesis of systemic lupus erythematosus. PMC4745266.
16. Cancer immune evasion through loss of MHC class I antigen presentation. *Frontiers in Immunology*. 2021. doi:10.3389/fimmu.2021.636568.
17. Variations in MHC class I antigen presentation and immunopeptidome selection pathways. PMC7525337.

Article 3: Mechanisms of Basic Immune Reactions

Immunology of the Cellular Clone

Abstract

Articles 1 and 2 of this series described which antigen categories exist and exactly where within the cell they are located. This article closes the next necessary layer: how the immune reactions directed at these antigens are themselves built, at the level of specific molecules and cascades rather than at the level of class labels. The article's task is purely mechanistic and context-independent: the analysis concerns the immune reaction as such, not how it is applied in particular cases (tumor, pregnancy, autoimmune disease) — those cases will make up the content of Article 4. The material is organized into four effector categories: innate mechanisms (NK cells, macrophages, complement, the inflammasome), adaptive cellular mechanisms (CD8+ cytotoxic T cells), humoral effector mechanisms (antibodies as an executive rather than a recognizing component), and the cytokine programs of T-helper polarization that determine which of the listed mechanisms will dominate in a given reaction.

1. Introduction

Classifying immune reactions (innate/adaptive, cellular/humoral) does not by itself explain why a target cell dies or survives — for that, one has to descend to the level of specific receptors, enzymes, and pores in the membrane. This article deliberately avoids discussing which reaction is “correct” or “effective” against a given type of clone — that is the subject of Article 4. Here only the mechanism is fixed: who recognizes what, through which receptor, what happens inside the target cell as a result, and how one mechanism differs from another structurally.

2. Innate effector mechanisms

2.1. NK cells: the balance of inhibitory and activating signals

[Established] NK-cell cytotoxicity is determined not by a single signal but by a balance between inhibitory and activating receptors, integrated in real time upon contact with each potential target. Inhibitory receptors — KIR (in humans) and the CD94/NKG2A heterodimer — recognize MHC class I molecules (classical HLA-A/B/C via KIR, the non-classical HLA-E via NKG2A) and signal through ITIM motifs in the cytoplasmic tail, suppressing activation.^[1,2]

The “missing self” hypothesis describes the consequence of this balance: an NK cell remains quiescent as long as the target displays sufficient MHC-I, but is activated when MHC-I expression is reduced or absent — a state typical of viral infection or tumor transformation. The activating receptor NKG2D, meanwhile, binds the ligands MICA/B and proteins of the ULBP family, which are normally absent on healthy cells and appear under cellular stress. Different KIR families recognize different HLA-C allotypes (KIR2DL1 — the C2 epitope, KIR2DL2/3 — the C1 epitope) and HLA-A/B carrying the Bw4 motif (KIR3DL1), which creates individual variability in the NK-cell repertoire between people.^[3]

The executive killing mechanism for an activated NK cell is granule exocytosis: perforin forms a pore in the target cell's membrane, through which granzymes (serine proteases) enter the

cytoplasm and trigger the effector caspase cascade, causing apoptosis of the target from within.^[1]

2.2. Macrophages: phagocytosis, the oxidative burst, and TNF- α secretion

[Established] Macrophages destroy a target through three parallel routes: direct phagocytosis (engulfment followed by lysosomal degradation), the oxidative burst (NADPH oxidase generates superoxide and derived reactive oxygen and nitrogen species inside the phagosome), and secretion of TNF- α , which can induce apoptosis in the target cell via the TNFR1 receptor independently of phagocytosis.

Note on classification (added on review): TNFR1-mediated apoptosis belongs to the same death-receptor superfamily architecture as Fas and the TRAIL receptors (Section 3) — death-domain recruitment of FADD and caspase-8 — and is folded into the “death-receptor apoptosis” category in the summary table and synthesis below, rather than left as an uncounted sixth mechanism.

Framework observation: a macrophage's decision to attack or hold back is governed by a separate set of inhibitory surface signals (“don't eat me”, including CD47–SIRP α , CD24–Siglec-10, MHC-I–LILRB1), already covered in detail in Framework 1.5 as clone-escape mechanisms; here these receptors are mentioned only as part of the normal physiological regulation of phagocytosis that prevents attacks on healthy cells, without re-examining their role in the tumor context.

2.3. Complement: three activation pathways, opsonization, the membrane attack complex

[Established] The three complement activation pathways converge on the same reaction — cleavage of C3 into C3a and C3b — despite different triggering mechanisms. The classical pathway is triggered by binding of C1q to antibodies already bound to the target surface. The lectin pathway is triggered by binding of mannose-binding lectin to carbohydrate structures on a pathogen's surface, and is structurally homologous to the classical pathway. The alternative pathway is activated by spontaneous C3 hydrolysis and requires no antibody, providing a baseline level of surveillance independent of adaptive immunity.^[4]

C3b, generated in all three pathways, functions as an opsonin — it coats the target surface and is recognized by complement receptors on phagocytes, sharply increasing the efficiency of phagocytosis. Further cleavage of C5 by the C5 convertase releases C5a (a potent mediator of inflammation and chemotaxis) and C5b, which initiates assembly of the membrane attack complex (MAC) from C6, C7, C8, and multiple copies of C9, forming a through-pore in the target's membrane and causing its osmotic lysis — a route to destruction that requires neither phagocytosis nor direct contact from an effector cell.^[4]

2.4. The inflammasome and pyroptosis

[Established] The NLRP3 inflammasome is a cytosolic multiprotein platform that activates the protease caspase-1 in response to a broad range of pathogen-associated and sterile danger signals. Active caspase-1 performs two parallel functions: proteolytic maturation of the pro-inflammatory cytokines IL-1 β and IL-18, and cleavage of the protein gasdermin D (GSDMD), releasing its N-terminal domain.^[10]

The N-terminal domain of GSDMD oligomerizes and inserts into the plasma membrane, forming stable pores — this is the executive stage of pyroptosis, a lytic form of cell death

distinct from apoptosis in that membrane integrity is not preserved until the very end of the process. Mature IL-1 β and IL-18 are released through these same pores, meaning cell death and the release of inflammatory mediators occur through a single structural mechanism rather than as two independent events.^[10]

3. Adaptive cellular mechanisms: CD8+ cytotoxic T lymphocytes

[Established] A CD8+ T cell recognizes not a whole antigen but a short peptide presented in complex with MHC class I, via the T-cell receptor (TCR) — unlike an NK cell, for which peptide specificity is not required. Once recognition occurs, the CTL employs two structurally independent mechanisms to destroy the target.

The first is the same granule-exocytosis mechanism used by NK cells: perforin forms a pore, and granzymes trigger apoptosis from within the target cell.^[1]

The second, structurally separate mechanism is the death-receptor pathway: binding of FasL (CD178) on the CTL surface to the Fas receptor (CD95) on the target causes Fas oligomerization and recruitment of the adaptor protein FADD via homotypic death-domain interaction; FADD, in turn, recruits procaspase-8, forming the DISC (death-inducing signaling complex). Active caspase-8 triggers effector caspases either directly (in “type I” cells) or through additional signal amplification via the mitochondrial pathway (in “type II” cells, where amplification through Bid and cytochrome c release is required for a sufficient apoptotic signal).^[5]

The TRAIL-receptor pathway (DR4/DR5) works in a structurally analogous way, through a set of receptors and ligands independent of Fas, and can be activated not only by CTLs but by NK cells as well — meaning receptor-mediated apoptosis is not an exclusively adaptive mechanism, but one that crosses the boundary between innate and adaptive immunity. TNFR1 (Section 2.2) is a third member of the same death-receptor superfamily, engaged by macrophage-secreted TNF- α rather than by a cell-surface ligand presented in trans.^[6]

Framework observation: the existence of two structurally independent killing pathways within the same effector cell (granule exocytosis and death receptors) means that a clone's escape from one pathway (for example, loss of sensitivity to perforin/granzymes through altered membrane composition) does not automatically protect it from the other — a clone must neutralize both independently, which has direct bearing on assessing the completeness of masking, to be addressed in later articles of this series.

4. Humoral effector mechanisms: the antibody as an executive component

An important structural observation for this section: an antibody by itself has no cytotoxic function — it acts as an adaptor, linking a specific antigen on the target to one of several independent effector mechanisms via its Fc domain.

[Established] Neutralization is the simplest mechanism: the antibody physically blocks a functional site on the target (for example, a virus's receptor-binding domain) without recruiting any additional effector.

Opsonization — the Fc domain of an antibody bound to the target is recognized by Fc receptors on phagocytes, sharply increasing the efficiency of phagocytosis through the same route as opsonization by complement C3b, but with different recognition specificity.

Complement-dependent cytotoxicity (CDC) — C1q of the classical complement pathway binds directly to the Fc domain of antibodies already anchored on the target surface, triggering the same cascade described in Section 2.3, all the way to MAC assembly.^[4]

Antibody-dependent cellular cytotoxicity (ADCC) — the receptor FcγRIIIa (CD16a) on the NK-cell surface binds the Fc domain of IgG coating the target, triggering formation of an immunological synapse and degranulation of the same perforin/granzyme granules used in direct NK cytotoxicity; alternatively, the NK cell may engage the same death receptors (Fas, TRAIL receptors) used by CTLs.^[7]

Framework observation: ADCC structurally represents a point of intersection between the humoral and innate mechanisms — recognition is mediated by an antibody (a product of adaptive immunity), but execution is carried out by the same molecular apparatus an NK cell uses for innate missing-self recognition. This means that, when classifying an immune reaction, it is more informative to separate the recognizing and executing components than to assign the reaction as a whole to a single category.

Summary table: the five executive killing mechanisms

Mechanism	Effector cell(s)	Key molecules	Mode of death
Membrane attack complex (MAC)	None required — complement alone	C5b, C6, C7, C8, C9 (§2.3)	Osmotic lysis
Granule exocytosis	NK cells, CD8+ CTLs, ADCC	Perforin, granzymes (§2.1, §3)	Caspase-driven apoptosis from within
Death-receptor apoptosis	CD8+ CTLs (Fas/FasL), NK cells (TRAIL), macrophages via secreted TNF-α (TNFR1)	Fas/FasL, DR4/DR5-TRAIL, TNFR1 — same death-domain/FADD/caspase-8 superfamily architecture (§2.2, §3)	Receptor-triggered apoptosis (DISC → caspase-8)
Pyroptosis	Any cell with an active inflammasome	NLRP3, caspase-1, gasdermin D (§2.4)	Lytic, pore-mediated; couples death to IL-1β/IL-18 release
Phagocytosis + oxidative burst	Macrophages	NADPH oxidase, lysosomal enzymes (§2.2)	Engulfment and degradation

5. Cytokine programs of T-helper polarization

CD4+ T helpers do not kill the target directly; their function is to determine which of the mechanisms described in Sections 2–4 will dominate in a given reaction, through secretion of polarizing cytokines.

[Established] The Th1 program is induced by IL-12 and produces IFN-γ and TNF-α — these cytokines enhance the macrophage oxidative burst (Section 2.2) and CD8+ T-cell cytotoxicity (Section 3), meaning Th1 reinforces cellular killing mechanisms.

The Th2 program is induced by IL-4, requires the transcription factor GATA3, and produces IL-4, IL-5, and IL-13 — a profile that supports a predominantly humoral response (Section 4) and an antiparasitic/allergic-type reaction rather than cellular cytotoxicity.^[8]

The Th17 program is induced by a combination of TGF- β , IL-6, IL-21, and IL-23 acting through the transcription factor ROR γ t, and produces IL-17A/F and IL-22 — a pro-inflammatory profile directed mainly against bacteria and fungi. IFN- γ (Th1) suppresses Th17 differentiation by enhancing T-bet, while IL-4 (Th2) suppresses it through an independent route — meaning the three programs are mutually antagonistic at the level of transcription factors, not merely independent of one another.^[8,9]

The Treg program is sustained by TGF- β and IL-2 acting through the transcription factor FoxP3, and produces IL-10 and TGF- β along with the inhibitory receptor CTLA-4 — this is active suppression of a reaction, not the absence of one: Treg cells can physically compete for IL-2 (depriving Th17 cells of the resource needed for their STAT5-mediated suppression) and directly suppress effector T cells through CTLA-4-mediated inhibition of costimulation.^[8]

Framework observation: Th-response polarization determines not the “strength” of an immune reaction but which of the structurally independent mechanisms (Sections 2–4) will be engaged — cellular cytotoxicity, a humoral response, or active suppression. This connects directly to the question left open in Article 2 (P1.13): if the type of dominant Th program systematically differs depending on the antigen category (embryonic/barrier/neoantigen/combinatorial) or its localization (surface/cytoplasmic/nuclear), this may mean the antigen's temporal and spatial axes are not fully independent of the immune reaction's outcome — a question to keep in mind when writing Article 4.

Summary table: Th-polarization programs

Program	Inducing cytokines	Master TF	Products	Dominant downstream mechanism
Th1	IL-12	<i>T-bet</i>	IFN- γ , TNF- α	Enhances macrophage oxidative burst and CD8+ cytotoxicity (cellular killing)
Th2	IL-4	<i>GATA3</i>	IL-4, IL-5, IL-13	Supports humoral response (§4); antiparasitic/allergic profile
Th17	TGF- β , IL-6, IL-21, IL-23	<i>RORγt</i>	IL-17A/F, IL-22	Pro-inflammatory, antibacterial/antifungal
Treg	TGF- β , IL-2	<i>FoxP3</i>	IL-10, TGF- β , CTLA-4	Active suppression — IL-2 competition and CTLA-4-mediated costimulation blockade

6. Synthesis

Five independent executive mechanisms of target-cell death have been described in this article: osmotic lysis via the complement membrane attack complex; granule exocytosis (perforin/granzymes), shared by NK cells, CD8+ T cells, and ADCC; receptor-mediated

apoptosis via the death-receptor superfamily (Fas/FasL and TRAIL, shared by CD8⁺ T cells and NK cells, plus TNFR1 engaged by macrophage-secreted TNF- α), grouped together as one mechanism despite three distinct receptor/ligand pairs, since all three converge on FADD/caspase-8; pyroptosis via caspase-1/gasdermin D, topologically distinct from apoptosis; and phagocytosis followed by lysosomal degradation, supplemented by the oxidative burst. None of these mechanisms is the exclusive property of innate or adaptive immunity — the recognizing and executing components can combine independently (most clearly seen in ADCC).

The Th1/Th2/Th17/Treg programs do not add a sixth killing mechanism; they redistribute weight among the five already described, depending on cytokine context. This map is the necessary mechanistic foundation for Article 4, which must show which of these mechanisms are actually engaged against the specific antigen categories described in Articles 1–2, and what happens when a clone neutralizes one mechanism but not the others.

Note: the labels [Established] / Framework observation correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4.

References

1. Wang K, Wang L, Wang Y, Xiao L, Wei J, Hu Y, Wang D, Huang H. Reprogramming natural killer cells for cancer therapy. *Mol Ther*. 2024;32(9):2835–2855. doi:10.1016/j.ymthe.2024.01.027.
2. NK cell self tolerance, responsiveness and missing self recognition. PMC3984600. (CD94/NKG2A recognition of HLA-E/Qa-1).
3. Immune selection during tumor checkpoint inhibition therapy paves way for NK-cell “missing self” recognition. PMC5537320. (KIR structure and HLA-C1/C2 specificity).
4. The complement cascade and renal disease. PMC3898353. (classical, lectin, and alternative pathways; C3b opsonization; membrane attack complex).
5. Strasser A, Jost PJ, Nagata S. The many roles of Fas receptor signaling in the immune system. *Immunity*. 2009;30(2):180–192. doi:10.1016/j.immuni.2009.01.001.
6. Kumar R, Herbert PE, Warrens AN. An introduction to death receptors in apoptosis. *Int J Surg*. 2005;3(4):268–277.
7. From CD16a biology to antibody-dependent cell-mediated cytotoxicity improvement. PMC9203678. (ADCC, Fc γ RIIIa/CD16a, perforin/granzyme and death-receptor pathways in ADCC).
8. Intricacies of TGF- β signaling in Treg and Th17 cell biology. PMC10468540. (Th1/Th2/Th17/Treg differentiation, cytokine profiles, FoxP3/ROR γ t).
9. The alterations in and the role of the Th17/Treg balance in metabolic diseases. *Frontiers in Immunology*. 2021. doi:10.3389/fimmu.2021.678355. (IFN- γ /IL-4 cross-inhibition of Th17 differentiation).
10. NLRP3 inflammasome and pyroptosis: implications in inflammation and multisystem disorders. PMC12360325. (NLRP3, caspase-1, gasdermin D, pyroptosis mechanism).

Article 4: The Cancer Case — Immune Dynamics of the Malignant Clone

Immunology of the Cellular Clone

Abstract

Articles 1–2 of this series described the clone's antigen categories and their localization; Article 3 described the mechanisms of basic immune reactions independent of context. This article applies that entire apparatus, for the first time, to one specific case — the malignant tumor — leaving other cases (pregnancy, autoimmune states) for separate future articles. The central organizing framework is the Elimination–Equilibrium–Escape model (immunoediting): the article shows which specific mechanisms from Article 3 are engaged in each of the three phases, and how the known masking mechanisms of the tumor clone (already described in detail in Framework 1.5) map one-to-one onto the mechanisms of Article 3 — each masking layer neutralizes one specific, named attack mechanism. The result is not a new catalog but a correspondence map of “attack → countermeasure,” which also refines the “double lock” hypothesis of dormancy proposed earlier in the framework.

1. Introduction

The tumor clone is not the only case, but historically the most thoroughly studied application of the antigenic architecture (Articles 1–2) and the immune-reaction mechanisms (Article 3). This article is deliberately limited to cancer: the combinatorial trophoblast model and the barrier model of immune privilege, also discussed in this series, find partial analogues in the tumor case (choriocarcinoma, testicular and CNS tumors respectively), but systematic application of the framework to pregnancy and autoimmune states as such remains the task of subsequent, separate articles.

2. Antigen categories in the tumor clone

2.1. Neoantigens

Established in Article 1 (Section 4) as a category with no pre-existing protection. Remains a central, though not the only, source of tumor antigen.

2.2. Oncofetal antigens: reactivation of the embryonic category

[Established] Alpha-fetoprotein (AFP) and carcinoembryonic antigen (CEA) are proteins normally expressed by the fetal yolk sac and liver, peaking around gestational weeks 12–16 and then falling silent entirely in the healthy adult tissue; their re-expression in hepatocellular carcinoma, colorectal cancer, and other tumors represents reactivation of precisely the gene that was silenced shortly after birth.^[4,6]

Framework observation: this is a direct intersection with the embryonic antigen category from Article 1 (Section 2) — but with the sign reversed. If an embryonic antigen is tolerized precisely because it is present in the body before the T-cell repertoire matures, an oncofetal antigen is a case where that same, already-tolerized protein from the embryonic window reappears decades later, in a context for which tolerance was never specifically established or re-established. Open question: does this preserve tolerance (the antigen is still “self”) or partially erode it with time — the literature gives grounds for the latter: the prototypical mouse

oncofetal antigen P1A can serve as a full-fledged target for cytotoxic T lymphocytes, meaning reactivation in a tumor context can restore immunogenicity that was lost for the same protein in the embryonic context.^[5]

If confirmed as a general pattern rather than a peculiarity of individual antigens, this would mean that tolerance established along the temporal axis (Article 1, P1.1–P1.2) is not a permanent property of the antigen but is tied to the pair “antigen + expression context” — a hypothesis requiring separate formalization in a future revision of the Article 2 postulates.

2.3. Combinatorial reactivation: choriocarcinoma

Already treated in Article 1 (Section 5) as a clone that inherited the trophoblast's combinatorial tolerance program. In the context of this article, choriocarcinoma matters as a demonstration that reactivation of a physiological program (Section 2.2 above) is not limited to individual embryonic antigens — an entire combinatorial tolerance system can be reactivated.

2.4. Localization of the tumor antigen

The surface/cytoplasm/nucleus axis (Article 1, Section 6) applies to the tumor cell without modification: apoptotic death of a tumor cell under therapy or immune pressure can expose nuclear material through the same route (blebbing) described for any cell; the difference is that during mass tumor cell death (for example, under chemotherapy), the volume of exposed material can exceed local clearance capacity, creating favorable conditions for secondary autoreactivity — a mechanism separate from the antitumor response itself and worth noting but not developing in this article.

3. The Elimination–Equilibrium–Escape model: three phases

3.1. Elimination: which mechanisms from Article 3 are engaged

[Established] The elimination phase engages several independent mechanisms from Article 3 simultaneously, not just one: CD8⁺ T cells recognize a neoantigenic peptide in complex with MHC-I and kill the target via granule exocytosis (perforin/granzymes) or via Fas/FasL; NK cells attack cells with reduced MHC-I expression via the missing-self mechanism; macrophages and complement participate in secondary clearance of dying cells. IFN- γ , produced by Th1-polarized CD4⁺ cells, enhances both CD8⁺ T-cell cytotoxicity and the macrophage oxidative burst — meaning elimination relies on the coordinated action of at least three structurally independent mechanisms from Article 3 at once, not on any single one.^[1,2]

3.2. Equilibrium: Darwinian selection, the Dormant Cell Pool, and two competing architectures of the “double lock”

[Established] In the equilibrium phase the immune system neither eliminates the clone entirely nor allows it to progress: prolonged lymphocyte and IFN- γ pressure acts as a selective force favoring genetically unstable, immunogenically “silent” variants of the clone, dependent on IL-12 and IL-23.^[1,3]

Framework interpretation: this is the immunological analogue of the Dormant Cell Pool, already described in the framework in energetic terms (the IGF1R–mTOR–NR2F1 axis). Combining the two lines refines the “double lock” hypothesis of dormancy — and here two distinct architectures need to be held apart as competing hypotheses, rather than collapsed into

one, until an experiment decides between them. The two variants are summarized in the table below.

Summary table: two competing architectures of the “double lock”

	Variant A — independent locks	Variant B — convergent inputs
Core claim	Energetic lock (IGF1R–mTOR–NR2F1 → H3K27me3) and immune lock (T/NK, phagocytosis, metabolic invisibility) are mechanistically independent; opening either is sufficient to exit dormancy.	Both pathways converge on one executive node — likely p27, induced by both NR2F1 and IFN-γ → IDO1-kynurenine → AhR; dormancy persists as long as at least one input is active.
Shared node?	No — no common executive molecule.	Yes — candidate: p27.
Exit condition	Either lock opening alone is sufficient.	Both inputs must be inactivated simultaneously.
Falsification criterion	Energetic and immune pathways shown to converge on the same node → Variant A false.	Clone with neutralized immune input (e.g., IFN-γR loss) remains dormant despite intact energetic branch (or vice versa) → favors independent nodes, against Variant B.
Test case prediction	Escape from one branch alone does not bring the clone closer to exiting dormancy.	Escape from one branch can be sufficient for exit if the other branch is already weakened.

Practical consequence of the distinction: the two variants make different predictions in a concrete, experimentally testable case: a clone with loss of IFN-γR alongside weak NR2F1 expression. Both variants are recorded here as equally live hypotheses — experiment, not argument, should decide between them; the framework does not yet commit to either.

3.3. Escape: a correspondence map of masking layers and neutralized mechanisms

Each masking layer previously described in Framework 1.5 corresponds to neutralization of one specific, named mechanism from Article 3 — this correspondence has not previously been assembled into an explicit table until now.

Summary table: masking layer → neutralized mechanism

Masking layer (molecules)	Neutralized mechanism	Article 3 reference
MHC-I / UPR downregulation (F1.5 layer 3.1)	Peptide-MHC-I recognition via the TCR — the central CD8+ CTL mechanism	§3 (<i>CD8+ CTLs</i>)
CD200–CD200R1; loss of ULBP (F1.5 layer 3.1)	NK-cell activation via missing-self/NKG2D	§2.1 (<i>NK cells</i>)
CD47–SIRPα, CD24–Siglec-10, MHC-I–LILRB1, PD-L1–PD-1, STC-1 (F1.5 layer 3.2)	Macrophage phagocytosis	§2.2 (<i>macrophages</i>)
Hypersialylation / Siglec-7/9 binding (F1.5 layer 3.3)	NK cytotoxicity — via a false inhibitory signal, distinct from ULBP loss	§2.1 (<i>NK cells</i>)
CD39/CD73/adenosine (F1.5 layer 3.4)	CD8+ T-cell and NK-cell function, post-	§2.1, §3

Masking layer (molecules)	Neutralized mechanism	Article 3 reference
	activation	
mCRPs: CD46, CD55, CD59 (new — seventh layer, this article)	Osmotic lysis via the membrane attack complex; CDC during antibody therapy	§2.3 (<i>complement</i>); §4 (<i>CDC</i>)
Antigen sink; antigenic modulation (Article 7)	Antibody-mediated mechanisms — opsonization, CDC, ADCC — via depletion of the effector pool	§4 (<i>humoral mechanisms</i>)
None identified	Pyroptosis (caspase-1/gasdermin D) — open gap, Section 4	§2.4 (<i>inflammasome</i>)

[Established] Membrane complement regulatory proteins (mCRPs) — CD46 (membrane cofactor protein), CD55 (decay-accelerating factor), and CD59 — are ubiquitously expressed to protect healthy tissue from excessive complement activation, but are systematically overexpressed in solid tumors as a separate escape mechanism: CD46 and CD55 limit C3b deposition and thus opsonization and classical-pathway activation, while CD59 directly blocks C9 polymerization and thereby assembly of the membrane attack complex (Article 3, Section 2.3), protecting the tumor cell from complement-dependent cytotoxicity during monoclonal antibody therapy as well (CDC, Article 3, Section 4).^[7,8]

Framework observation: mCRPs constitute a seventh independent masking layer not previously included in the Framework 1.5 inventory (which described the T/NK layer, the phagocytosis layer, the glycocalyx, and the metabolic layer, but not complement-specific protection); it should be added as a separate node in the next revision of Framework 1.5's structure, since it neutralizes a mechanism — osmotic lysis via the MAC — not covered by any of the four previously described layers.

Antigen sink and antigenic modulation (Article 7, “Active tactics for reducing immune pressure”) neutralize the antibody-mediated mechanisms of Article 3 (Section 4) — opsonization, CDC, and ADCC — by depleting the circulating antibody pool or its specific availability, rather than by blocking the executive mechanism at the target itself, as mCRPs do.

4. An open gap in the correspondence map: the inflammasome and pyroptosis

None of the known, documented tumor-escape mechanisms systematically neutralizes pyroptosis (Article 3, Section 2.4) the way mCRPs neutralize complement or phagocytosis checkpoints neutralize macrophage clearance. This could mean one of three things: (a) pyroptosis is not a significant surveillance mechanism against the tumor clone specifically under natural conditions, unlike the infectious context for which it is primarily documented; (b) a corresponding escape mechanism exists but has not yet been systematically described in the available literature; (c) the absence of a symmetric countermeasure leaves the tumor with a residual vulnerability, potentially usable therapeutically (inducing pyroptosis as a strategy that bypasses escape mechanisms the tumor has already developed against apoptosis). Distinguishing between these three options is a task for future revision, not for this article; here only the fact of the asymmetry in the correspondence map is recorded.

5. Synthesis

The tumor clone does not invent antigen categories anew (Section 2 shows it either generates mutagenic noise or reactivates one of the already-existing physiological programs — embryonic or combinatorial), and it does not invent new mechanisms of destruction or defense (Section 3 shows that Escape is fully described by neutralization of specific mechanisms already catalogued in Article 3). The only genuinely new element introduced by the tumor context itself is Equilibrium as a phase of active Darwinian selection — and, depending on which variant of Section 3.2 holds, this selection may be integrated with the energetic lock of dormancy into a single convergent process (Variant B) rather than a merely parallel one (Variant A).

The correspondence map “masking layer → neutralized mechanism” (Section 3.3), supplemented by a seventh layer (mCRPs) and an explicitly recorded gap regarding pyroptosis (Section 4), gives the framework something Framework 1.5 did not have at its initial assembly: a complete, as far as possible exhaustive rather than illustrative map, in which a specific countermeasure is identified (or noted as absent) for every known attack mechanism. This is the operational result of applying Articles 1–3 to a concrete case, as promised at the start of this series.

Note: the labels [Established] / Framework observation / Framework interpretation correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4. Full bibliographic data for the Framework 1.5 masking layers (phagocytosis checkpoints, glycocalyx, the metabolic axis) are given in the original Framework 1.5 document and are not duplicated here; only sources newly introduced in this article are listed below.

References

1. Mittal D, Gubin MM, Schreiber RD, Smyth MJ. New insights into cancer immunoediting and its three component phases — elimination, equilibrium and escape. *Curr Opin Immunol.* 2014;27(1):16–25. doi:10.1016/j.coi.2014.01.004.
2. Dunn GP, Bruce AT, Ikeda H, Old LJ, Schreiber RD. Cancer immunoediting: from immunosurveillance to tumor escape. *Nat Immunol.* 2002;3(11):991–998. [Corrected on review: the foundational paper defining the elimination–equilibrium–escape model was present in the reference list but not cited anywhere in the text; now cited alongside ref. 1 in Section 3.1.]
3. The senescence journey in cancer immunoediting. PMC10983694. PMID 38561826.
4. Alpha-fetoprotein: a multifaceted player in cancer biology. PMC12288575.
5. Transgenic expression of P1A induced thymic tumor: a role for onco-fetal antigens in tumorigenesis. PMC2955541. [Corrected on review: this source supports the P1A/CTL claim in Section 2.2 but was not cited there in the original draft.]
6. Overview of oncofetal antigens in cancer. *Ann Clin Lab Sci.* 1983;13(4):261.
7. The role of membrane bound complement regulatory proteins in tumor development and cancer immunotherapy. *Frontiers in Immunology.* 2019. doi:10.3389/fimmu.2019.01074.
8. The complement receptors CD46, CD55 and CD59 are regulated by the tumour microenvironment of head and neck cancer to facilitate escape of complement attack. *PubMed* 24915776.

Article 5: The Pregnancy Case — Immune Tolerance at the Maternal–Fetal Interface

Immunology of the Cellular Clone

Abstract

Article 4 applied the antigenic architecture (Articles 1–2) and the mechanisms of immune reaction (Article 3) to the tumor clone. This article applies the same apparatus to a second natural case of tolerance toward an antigenically foreign tissue — pregnancy, where the trophoblast, already described in Article 1 (Section 5) as a combinatorial model, is treated not as a source of antigens but as a masking system operating in real time. Nearly every escape mechanism documented for the tumor in Article 4 has a direct, and in some cases historically earlier, counterpart at the maternal–fetal interface — for IDO and the complement regulatory proteins, pregnancy was in fact the first context in which the mechanism was discovered, with the transfer to oncology occurring later. The key divergence from the tumor case is examined critically: does maternal–fetal tolerance, sustained without cell-cycle arrest, tell us anything about the independence of the energetic and immune locks proposed in Article 4? The analysis in Section 5 has been revised twice — first to narrow an overreaching claim, and again to surface an unresolved internal tension between that narrowed claim and the IDO mechanism described in Section 3.2, along with an evidentiary gap regarding p27 status in extravillous trophoblast that the hypothesis space in Section 5 depends on but does not itself establish.

1. Introduction

The trophoblast is not merely an illustration of an antigen category from Article 1 but a full test case for the entire apparatus of Article 3: here an antigenically foreign (semi-allogeneic) tissue is neither destroyed nor placed into dormancy in the strict sense, but maintained in a tolerant state for a fixed, predetermined interval. This distinguishes pregnancy from the tumor case already at the level of the question being asked: where Article 4 asked “how does the clone escape and progress,” the question here is “how is a tissue tolerated and then released from the need for tolerance.”

2. Antigenic basis: the combinatorial model in action

The combinatorial HLA-G/E/F program and the absence of HLA class II on the trophoblast were already established in Article 1 (Section 5) as the physiological source of tolerance later inherited by choriocarcinoma. In this article, that program is treated not as a static antigenic profile but as an entry point for specific cellular escape mechanisms, each of which must be separately mapped onto the catalog in Article 3 — exactly as was done for the tumor in Article 4.

3. Correspondence map: layers of pregnancy tolerance and the mechanisms they neutralize

3.1. The NK layer: decidual NK cells

[Established] More than 70% of decidual leukocytes in the first trimester are NK cells with a CD56bright/CD16-negative phenotype — a subset that makes up only about 5% of peripheral

blood NK cells but dominates precisely at the site of contact with the trophoblast. This subset is characterized by few cytotoxic granules and low cytotoxicity compared with the CD56dim/CD16-positive subset that dominates the periphery and accounts for most NK-mediated killing (Article 3, Section 2.1).^[1,2]

In addition to the intrinsic low cytotoxicity of this subset, decidual NK cells interact directly with HLA-G, HLA-C, and HLA-E ligands expressed by extravillous trophoblast, further suppressing their lytic capacity — meaning the NK-tolerance layer here consists of two independent components, an altered cell subset and direct receptor inhibition, rather than a single mechanism.^[3]

Framework observation: a closely analogous polarization toward the CD56bright/CD16-negative, low-cytotoxicity, predominantly cytokine-producing phenotype has been proposed as a hypothesis in tumor-infiltrating NK cells and in circulating NK cells of patients with advanced malignancies, where it correlates with an immunosuppressive microenvironment. The source of this parallel frames it explicitly as a hypothesis to be tested (“it is possible to hypothesize that the mechanisms... operate in a similar way”), not as a demonstrated identity of mechanism — the wording here has been corrected accordingly from an earlier draft that overstated this as an established identity. With that caveat, the NK-layer mechanism in Article 4 (Framework 1.5, layer 3.1: loss of ULBP, CD200–CD200R1) and the NK-tolerance mechanism in pregnancy are worth recording as a plausibly general, structurally related phenomenon rather than one isolated to each case.^[9]

3.2. The metabolic layer: IDO/kynurenine — a case where the direction of discovery was reversed

[Established] A classic experiment showed that pharmacological inhibition of indoleamine 2,3-dioxygenase (IDO) in pregnant mice with 1-methyl-tryptophan caused rapid T-cell-mediated rejection of allogeneic conceptuses, while syngeneic conceptuses were unaffected — meaning that local tryptophan depletion by the trophoblast specifically, and not some other mechanism, was necessary to prevent rejection of the allogeneic fetus.^[4]

Framework observation: this is the historically earlier source of a mechanism already incorporated into Framework 1.3/1.5 as the IFN- γ → IDO1-kynurenine → AhR → p27 axis explaining immune dormancy of tumor cells: this same axis was first documented in the context of pregnancy in 1998, with the transfer to oncology occurring later. The order of discovery here is the reverse of most other correspondences in this series, where the mechanism was typically described for the tumor first.

An important caveat against oversimplification: subsequent work with IDO1-knockout mice did not show rejection of allogeneic conceptuses, which calls into question the sufficiency of IDO1 alone and points to a compensatory role for the related enzyme TDO (tryptophan 2,3-dioxygenase), expressed by decidual stromal cells with a peak around day 6.5 of murine gestation — meaning the metabolic tolerance layer is redundant and not reducible to a single enzyme, which matters when assessing the robustness of the analogous mechanism in the tumor context.^[5,6]

Tissue-specificity note (added on review): the evidence that IDO induction in trophoblast is IFN- γ -dependent comes from villous explant cultures — that is, from villous cytotrophoblast/syncytiotrophoblast, not from extravillous trophoblast (EVT) specifically. Section 5 below builds an argument specifically about EVT. The two populations are not interchangeable (Section 5.1), so this mechanism cannot be assumed, without further evidence,

to hold in the same form in EVT. This mismatch is carried forward explicitly into Section 5 rather than left implicit.^[5]

3.3. The complement-regulatory layer: the same protein set as Article 4's seventh layer

[Established] The membrane complement regulatory proteins CD46 (membrane cofactor protein) and CD55 (decay-accelerating factor) are widely expressed on syncytiotrophoblast and endovascular trophoblast from at least gestational week 6 to term; CD59, which blocks C9 polymerization and thus assembly of the membrane attack complex (Article 3, Section 2.3), is expressed especially highly on endovascular trophoblast, which invades maternal spiral arteries and is in direct contact with maternal blood.^[7,8]

In mice, deletion of the rodent-specific regulatory protein Crry (a functional analogue of human CD46/CD55) causes spontaneous complement deposition in the placenta at the maternal–fetal interface and embryonic lethality — direct experimental evidence that this layer is not redundant but necessary, in contrast to the metabolic layer (Section 3.2), where functional redundancy was demonstrated.^[10]

Framework observation: this is the same set of molecules (CD46/CD55/CD59) identified in Article 4 as a seventh masking layer, previously absent from the Framework 1.5 inventory, protecting the tumor clone from complement-mediated lysis. Moreover, characterization of these proteins on the trophoblast (for example, work from the early 1990s on the distribution of CD59 in placenta) historically precedes the systematic description of their overexpression in solid tumors, documented later. This means the seventh layer from Article 4 is more accurately viewed not as an oncology-specific discovery but as the transfer of an already-known reproductive-physiology mechanism into a different context — the same logic of inheriting a physiological program already applied to choriocarcinoma in Article 1.

3.4. The Treg layer

The trophoblast produces thymic stromal lymphopoietin (TSLP), which shifts dendritic cells toward a Th2 profile and expands regulatory T cells in the decidua; additional immunoregulatory molecules, including IL-10 and galectin-9, contribute to the same Treg-mediated suppression layer already described mechanistically in Article 3 (Section 5) as active, rather than passive, suppression of the response.

Summary table: pregnancy–tumor correspondence

Layer	Pregnancy mechanism (§3)	Tumor counterpart (Article 4)	Direction of discovery
NK (§3.1)	CD56bright/CD16 [−] decidual NK, low cytotoxicity + HLA-G/C/E receptor inhibition	Loss of ULBP, CD200–CD200R1 (F1.5 layer 3.1)	<i>Proposed as hypothesis in tumor NK cells (not established identity)</i>
Metabolic (§3.2)	IDO/kynurenine — tryptophan depletion blocks T-cell rejection of allogeneic conceptus	IFN-γ → IDO1-kynurenine → AhR → p27 axis (F1.3/1.5)	<i>Pregnancy first (1998); oncology later</i>
Complement-regulatory (§3.3)	CD46/CD55/CD59 on syncytio-/endovascular trophoblast;	mCRPs — seventh masking layer (Article 4, §3.3)	<i>Pregnancy first (early 1990s);</i>

Layer	Pregnancy mechanism (§3)	Tumor counterpart (Article 4)	Direction of discovery
	Crry essential in mice		<i>oncology later</i>
Treg (§3.4)	TSLP-driven Th2 shift, decidual Treg expansion, IL-10/galectin-9	Treg-mediated suppression (Article 3, §5)	<i>Not established which context is earlier</i>

4. Immunoediting outside oncology: a formal extension of the model

[Established] The three-phase Elimination–Equilibrium–Escape model, developed to describe tumor immunity, has since been explicitly extended to maternal–fetal tolerance as a unifying framework explaining implantation, maintenance of pregnancy, and obstetric pathology (preeclampsia, recurrent pregnancy loss, preterm birth) as failures of specific immune phases rather than isolated pathologies.^[11]

Framework observation: the transfer of the model is bidirectional: the same conceptual apparatus that describes tumor immune evasion has independently been proposed to explain immune escape in endometrial cancer through direct comparison with tolerance mechanisms at the maternal–fetal interface — meaning the direction of the analogy in the literature already runs both ways (pregnancy → cancer and cancer → pregnancy), not one way, which supports the view that this parallel is not specific to the present framework but reflects a genuine commonality of mechanism, independently documented by different research groups.^[12]

5. Pregnancy as a partial test case for the “double lock” hypothesis — revised

This section has been revised twice. The first revision narrowed an overreaching claim (below, 5.1–5.3). The second revision, made on critical re-review, surfaces two problems in the narrowed version itself: an internal tension with Section 3.2, and a load-bearing evidentiary gap. Both are recorded rather than smoothed over.

5.1. First correction: which trophoblast, specifically

The trophoblast is not a single population. Syncytiotrophoblast, the barrier layer in direct contact with maternal blood, is terminally differentiated and post-mitotic by nature, formed by fusion of mononuclear cells that genuinely exit the cell cycle. Extravillous trophoblast (EVT), by contrast — the population that invades the spiral arteries and carries HLA-G — proliferates and migrates. The argument in this section concerns EVT specifically, not “the trophoblast” as an undifferentiated whole; treating the two populations as one would let a critic point to the arrested syncytiotrophoblast as a counterexample.

5.2. Second correction: absence of p27 arrest is not the same as absence of immune-branch input — and this hedge is itself in tension with Section 3.2

The argument only holds if IFN-γ signaling actually reaches EVT and proceeds normally but simply fails to induce p27. If, instead, IFN-γ signaling to the trophoblast is itself dampened upstream, then what is observed would be “input never arrived,” not “node unresponsive,” and the case would not test what it appears to test.

Tension noted on review: this hedge is difficult to reconcile with Section 3.2’s own evidence, which shows IDO expression in trophoblast is IFN-γ-inducible — meaning some functional

IFN- γ receptor signaling must be reaching the tissue in order to transactivate IDO in the first place. Taken at face value, this undercuts the hedge and reopens support for the stronger, pre-revision version of the argument. The tension is not fully resolved here for a specific reason: the IFN- γ /IDO evidence (Section 3.2) comes from villous trophoblast explants, while this section concerns EVT specifically (Section 5.1); whether IFN- γ receptor signaling is similarly intact and functional in EVT, as opposed to villous trophoblast, has not been established by any source cited in this article. The honest position is therefore that the hedge is neither confirmed nor refuted by the evidence assembled here — it depends on a tissue-specific fact (IFN- γ pathway activity in EVT specifically) that remains unmeasured in the literature surveyed for this article.^[5]

This requires a specific, currently unverified fact: whether active IFN- γ signaling (downstream STAT1/AhR activation, not merely cytokine presence) reaches EVT at a level comparable to that seen in tumor dormancy, while p27 induction is nonetheless absent. Until this is established, the argument below is a plausible hypothesis, not a demonstrated fact, and is labeled accordingly.

5.3. Third correction: a missing third architecture — and a gap in what it assumes

Article 4 posed a binary choice: two independent locks (Variant A) versus full, unconditional convergence on a shared node such as p27 (Variant B). A third, plausibly more realistic architecture was omitted from Article 4.

Variant C — conditional convergence. The executive node is molecularly shared, but its activation by the immune-branch input is conditional on a pre-existing cellular context — for example, prior NR2F1 expression or a corresponding chromatin state — which EVT simply lacks. Under Variant C, IFN- γ $\rightarrow$ IDO $\rightarrow$ AhR signaling can induce p27 in a tumor clone that already carries the right epigenetic groundwork, yet fail to arrest a proliferating EVT that does not, without requiring two separate nodes (Variant A) or unconditional convergence (Variant B). The difference is not whether a shared node exists, but whether the cell is currently competent to respond to it.

Evidentiary gap noted on review: no source cited anywhere in this article reports whether p27 is expressed, absent, or epigenetically silenced in EVT specifically. The entire choice among Variants A, B, and C in this section is therefore a discussion of which architecture would best explain a fact pattern — immune tolerance coexisting with proliferation — without direct measurement of the one molecule (p27) the discussion is organized around. This does not invalidate the reasoning, but it means Section 5's conclusion (5.4) should be read as identifying which hypothesis is most parsimonious given indirect evidence, not as one supported by direct data on p27 status in EVT.

Summary table: three variants of the “double lock” hypothesis

	Variant A — independent locks	Variant B — unconditional convergence	Variant C — conditional convergence
Core claim	Energetic and immune locks share no executive node.	Both inputs converge unconditionally on one node (p27).	Node is shared, but activation depends on prior cell state (e.g., NR2F1/chromatin).
Explains EVT	Yes, directly.	No — predicts arrest whenever immune input is present.	Yes — EVT lacks the prerequisite state, so the

	Variant A — independent locks	Variant B — unconditional convergence	Variant C — conditional convergence
proliferation + tolerance?			node doesn't fire despite input.
What pregnancy data show (this article)	Not established — requires ruling out tissue-specific IFN- γ pathway gaps in EVT.	Weakened by EVT proliferation, but not disproven without direct p27 data.	Most parsimonious given indirect evidence, but not demonstrated — p27 status in EVT unmeasured.
Falsification criterion	Energetic/immune pathways shown to converge on one node $\rightarrow$ false.	Clone with neutralized immune input stays dormant despite intact energetic branch $\rightarrow$ false.	p27 induction found independent of NR2F1/chromatin status $\rightarrow$ collapses into A or B depending on direction of result.

Falsification criterion (clarified): if direct measurement shows p27 is induced by the IFN- γ axis in EVT regardless of NR2F1/chromatin status (i.e., the node fires whenever the input is present, with no dependence on prior cell state), this rules out Variant C in favor of Variant B (unconditional convergence) combined with some other explanation for the lack of arrest. If p27 is not induced in EVT under IFN- γ exposure sufficient to induce it in a tumor clone, and no cell-state-dependent explanation accounts for the difference, this favors Variant A (fully independent nodes) instead. Variant C specifically requires showing that the same input produces different p27 outcomes in different cell states — the intermediate, context-dependent result distinguishing it from both A and B.

5.4. Revised conclusion

[Framework hypothesis] given the tension noted in 5.2 and the evidentiary gap noted in 5.3, the coexistence of immune tolerance with active EVT proliferation is compatible with all three variants, and Variant C (conditional convergence) is offered as the most parsimonious of the three — but this is a judgment about parsimony in the absence of direct data, not a conclusion drawn from it. Pregnancy does not establish independence of the two locks, nor does it cleanly rule out convergence; the honest summary is that the pregnancy case, as currently evidenced, underdetermines the choice among all three architectures, with Variant C carrying the lightest assumptions.

This is substantially weaker than the claim first proposed in this series, and weaker again than the first revision of it — each round of scrutiny has narrowed what can be said. Resolving this requires two specific measurements neither this article nor its sources provide: IFN- γ pathway activity (not just cytokine presence) in EVT specifically, and p27 status in EVT under that exposure. Both are named here as concrete experimental targets rather than left as vague calls for “further research.”

6. A key divergence from the tumor case: boundedness in time

Tolerance of the trophoblast is strictly bounded by the duration of pregnancy and resolves at parturition — none of the layers in Sections 3.1–3.4 becomes a permanent feature of the maternal organism. This fundamentally distinguishes the pregnancy case from Escape in the tumor model (Article 4, Section 3.3), where the same categories of mechanism (loss of MHC-I,

NK polarization, metabolic suppression, complement regulation) become fixed in the clone irreversibly and progressively intensify rather than resolve. Framework hypothesis: the same set of physiological tolerance tools produces a healthy outcome when engaged temporarily (successful pregnancy) and a pathological one when engaged permanently and progressively (tumor Escape); the boundary between the two outcomes is set not by which mechanisms are engaged but by their reversibility and duration.

Falsification criterion: *if temporary and permanent versions of these mechanisms are shown to be governed by fundamentally different molecular switches, rather than by the degree or duration of activation of the same ones, the hypothesis requires revision. Note that, as in Section 5, this section proposes no direct evidence of its own; it is flagged as the most speculative claim in the article and should be weighted accordingly relative to Sections 3.1–3.4, which rest on established, cited mechanisms.*

7. Synthesis

Nearly all the masking layers catalogued for the tumor in Article 4 have direct counterparts at the maternal–fetal interface, and for the metabolic (IDO) and complement-regulatory (mCRP) layers, pregnancy was historically the first context of discovery rather than a secondary application of the oncology model. This reinforces the thesis of Article 4 that the clone does not invent escape mechanisms anew but reactivates already-existing physiological programs — with the added finding here that at least two of these layers were themselves first described not in a tumor but in a reproductive context.

The main analytical contribution of this article, after two rounds of revision, is more modest than either version first proposed: pregnancy does not resolve the choice between Variant A and Variant B from Article 4, and on closer inspection does not even cleanly motivate Variant C as demonstrated rather than merely plausible — the argument depends on a fact (p27 status in EVT) that has not been measured, and contains an internal tension (Section 5.2) between the hedge it relies on and the mechanism established in Section 3.2. The question remains open for direct testing, and this article's contribution is to specify exactly what that test would need to measure, rather than to claim it has already been answered.

Note: the labels [Established] / Framework observation / Framework interpretation / Framework hypothesis correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4.

References

1. CD56 antigen — decidual NK cell phenotype and function. ScienceDirect Topics overview. [Secondary/tertiary source — curated summary, not peer-reviewed primary literature; cited only for the basic phenotype description, corroborated by ref. 2.]
2. TGFβ promotes conversion of CD16+ peripheral blood NK cells into CD16– NK cells with similarities to decidual NK cells. Proc Natl Acad Sci USA. 2007.
3. Role of decidual natural killer cells in human pregnancy and related pregnancy complications. PMC8426434.
4. Munn DH, Zhou M, Attwood JT, Bondarev I, Conway SJ, Marshall B, Brown C, Mellor AL. Prevention of allogeneic fetal rejection by tryptophan catabolism. Science. 1998;281(5380):1191–1193. doi:10.1126/science.281.5380.1191.

5. Tryptophan degradation by human placental indoleamine 2,3-dioxygenase regulates lymphocyte proliferation. PMC2288791. [Villous explant data; not EVT-specific — see Sections 3.2 and 5.2.]
6. Tryptophan 2,3-dioxygenase expression identified in murine decidual stromal cells is not essential for feto-maternal tolerance. PMC7752949.
7. Endovascular trophoblast expresses CD59 to evade complement-dependent cytotoxicity. ScienceDirect. 2019.
8. Complement regulatory proteins at the feto-maternal interface during human placental development: distribution of CD59 by comparison with membrane cofactor protein (CD46) and decay accelerating factor (CD55). PubMed 1376264.
9. Decidual-like NK cell polarization: from cancer killing to cancer nurturing. Cancer Discovery, AACR. 2021. [Frames the pregnancy–cancer NK parallel explicitly as a hypothesis, not an established identity — see Section 3.1.]
10. Xu C, Mao D, Holers VM, Palanca B, Cheng AM, Molina H. A critical role for murine complement regulator Crry in fetomaternal tolerance. Science. 2000;287(5452):498–501. doi:10.1126/science.287.5452.498.
11. Immunoediting in pregnancy: a new paradigm for understanding fetal tolerance and obstetric disease. J Perinat Med. 2026. doi:10.1515/jpm-2025-0246.
12. Endometrial cancer immune escape mechanisms: let us learn from the fetal–maternal interface. Frontiers in Oncology. 2020. PMC7080858.

Article 6: Boundaries of Applicability — Failure, Borrowing, and a Case Outside the Architecture

Immunology of the Cellular Clone

Abstract

Articles 4–5 applied the antigenic architecture and immune-reaction mechanisms to two cases in which adaptive immunity plays a central role — the tumor and pregnancy. This article tests the durability of that architecture against four additional variants of immune response: three of them (autoimmunity, transplantation, chronic infection), treated briefly in Part I, confirm the architecture — either by contradiction (failure of an already-described layer), by direct engineering borrowing, or by a pattern of reversed direction of scientific discovery. The fourth (regeneration), treated in depth in Part II, is a probable case lying entirely outside the antigenic architecture of Article 1, rather than another application of it.

Introduction

Building a durable model requires more than testing the architecture on one or two cases where it works well — it requires deliberately analyzing other variants of immune response as well, including ones where the architecture's mechanism breaks down (autoimmunity), where it is absent and must be artificially recreated (transplantation), where it is independently discovered in a different context (chronic infection), and, most importantly for the model's durability, the variant where the very question of antigenic architecture may not apply at all (regeneration). Such testing against heterogeneous material is not the accumulation of additional illustrations but a stress test: if an architecture only explains the cases it was built to explain, it has limited value; if it also predictably describes its own failures, its borrowings, and — the strictest test of all — its own boundaries of applicability, this increases confidence that it is correct.

Part I of this article treats three cases briefly, since each is fully described by the categories already established in Articles 1–5 without requiring an extension of the apparatus. Part II treats regeneration in depth, since it is the one of the four cases where preliminary analysis points toward stepping outside the architecture rather than applying it again — and such a conclusion requires more careful justification than a confirming case does.

Part I. Three cases of failure and borrowing

1. Autoimmune diseases

[Established] An autoimmune disease does not require a separate, sixth mechanism — it is the failure of one of the three already-described tolerance layers, running in reverse.

The first route to failure is loss of physical sequestration in the barrier antigen category (Article 1, Section 3): breach of the blood–testis barrier causes autoimmune orchitis, breach of the ocular barrier causes sympathetic ophthalmia. Here tolerance never existed in the first place — the antigen simply was never presented, and when it appears, the response proceeds as though against genuinely foreign material.

The second route is a clearance mechanism that fails to keep pace with exposure of material (Article 1, Section 6): a defect in the removal of apoptotic blebs leads to production of

antinuclear antibodies in systemic lupus erythematosus. Here the same physical process (blebbing), normally non-immunogenic, becomes immunogenic solely because of inefficient clearance.

The third route is a suppressive Treg layer (Article 3, Section 5) that is quantitatively or functionally insufficient, so that autoreactive T-cell clones already tolerated, and not removed by central selection, go unchecked at the periphery.

Framework observation: all three routes are not the emergence of a new mechanism but the breakdown of one of three already-documented ones: barrier, clearance, or suppression. An autoimmune disease, in this sense, is diagnosed by which of the three layers has failed, rather than by belonging to a new category of pathology.

2. Transplantation

[Established] HLA-G — the same molecule that provides trophoblast tolerance (Article 1, Section 5; Article 5) — is currently being tested directly as a therapeutic agent for inducing tolerance in transplantation: recombinant HLA-G5 significantly reduced histologically confirmed acute rejection in experimental small bowel transplantation, and tissue expression of HLA-G correlates with favorable transplant outcomes in humans, acting through the same inhibitory receptors ILT-2/ILT-4 on T cells, B cells, and NK cells already described for the trophoblast.^[1,2]

Framework observation: this is not an analogy but a direct borrowing of the same molecular tool from one physiological case into another, engineered one. An ordinary transplant is rejected precisely because it does not carry the combinatorial HLA-G program — unlike the trophoblast and choriocarcinoma, which either create it physiologically or inherit it. Efforts to induce tolerance in transplantation are efforts to artificially reproduce, on a foreign organ, what the trophoblast does on its own.

3. Chronic infections

[Established] T-cell exhaustion — a state of CD8+ T-cell dysfunction with progressive loss of proliferative capacity, effector function, and long-term persistence — was first systematically defined in the model of chronic LCMV infection, where PD-1 was established as the principal regulator of exhaustion, and reversibility of the dysfunction via PD-1/PD-L1 blockade was demonstrated there, well before the same findings were transferred to oncology.^[3,4]

Framework observation: this is the third case in this series of reversed direction of discovery — following IDO (Article 5, Section 3.2) and the complement regulatory proteins (Article 5, Section 3.3), both first documented in the context of pregnancy rather than cancer. A coincidence across three independent cases no longer looks accidental: the immune system appears to reuse the same limited toolkit of tolerance and exhaustion for any sustained antigenic contact, whether fetus, virus, or tumor. The order of discovery was determined not by the logic of the subject matter but by which experimental model happened to be technically more convenient in a given decade: LCMV and mouse pregnancy were studied earlier and more easily than the analogous processes in tumors.

Interim summary of Part I

None of the three cases adds a new layer to the architecture established in Articles 1–5. Autoimmunity confirms the architecture by contradiction, transplantation confirms it directly

through engineered borrowing, and chronic infection confirms the reversed-discovery pattern already noted in Article 5, bringing the count of such cases to three. Part II turns to the fourth case, where preliminary analysis points to a fundamentally different result.

Part II. Regeneration — a case outside the architecture

4. Innate immunity as a necessary and sufficient condition

[Established] Depletion of macrophages in the axolotl completely blocks limb regeneration, turning the amputated tissue into a fibrotic scar with epithelial hyperplasia instead of forming a blastema — meaning macrophages are not merely conducive to regeneration but strictly necessary for it.^[5,7,8]

Adaptive immunity, by contrast, is not required for regeneration: Rag1-knockout axolotls and zebrafish, completely lacking mature T and B cells, regenerate limbs, tails, and fins with normal kinetics, patterning, and skeletal outcome — the absence of lymphocytes is compensated by heightened innate (myeloid) activity in the blastema.^[6]

Framework observation: this is a direct inversion of the hierarchy established for cancer (Article 4) and pregnancy (Article 5), where the adaptive branch (CD8+ T cells, Th polarization, antibodies) was structurally central and innate mechanisms were auxiliary or supplementary. For regeneration, the hierarchy is reversed: innate clearance is necessary and sufficient, and adaptive immunity is dispensable altogether.

5. Same organism, same cell, different outcome: the neonatal mouse heart

[Established] Myocardial infarction in a newborn mouse (day P1) leads to full regeneration of the cardiac muscle in a macrophage-dependent manner, whereas an identical injury just 6 days later (day P7) leads to fibrotic scarring driven by pro-fibrotic macrophages. Depletion of macrophages in neonatal mice blocks regeneration and neoangiogenesis, producing the same fibrotic scars seen in adults.^[9,10]

The key difference is not the presence of macrophages (they are present in both cases) but their polarization profile and transcriptional program: regenerative P1 macrophages secrete factors that promote angiogenesis and new myocardium formation, whereas macrophages from older tissue produce soluble factors that stimulate fibrosis. Clearance of the macrophages themselves through the maturing cardiac lymphatic system also differs between P1 and P7 — at P1 macrophages are retained in the tissue longer, while at P7 they are cleared more quickly via the lymphatics to the mediastinal lymph nodes.^[11,12]

Framework observation: this is a controlled natural experiment in the same organism, the same cell lineage (macrophages), with the same type of injury — the single variable separating regeneration from scarring is the time window and its associated polarization program. This is structurally identical to the logic already established in Article 1, Section 6 for apoptotic blebs: not the nature of the event (cell death / organ injury) but the controllability and timeliness of clearance determines whether the outcome remains “silent” (regeneration) or becomes pathological (autoimmune reaction / fibrosis).

6. Framework hypothesis: regeneration as a control case outside antigenic architecture

If correct, regeneration is not a sixth case of applying the categories of Article 1 but a case where the very question of antigenic architecture does not apply: self tissue was never foreign, barrier-sequestered, combinatorial, or mutational in the sense of Article 1's sections — the problem of antigen tolerance is simply absent here. The only variable in play is the quality and balance of clearance of debris and senescent cells — that is, a narrow innate layer (Article 3, Section 2.2: phagocytosis, oxidative burst, but without “don't eat me” phagocytosis checkpoints, since there is no target that needs to be hidden).

Falsification criterion: if regenerating tissue is shown to express any of the large-scale masking layers described for the tumor and pregnancy (Article 4, Section 3.3) — for example, significant activation of phagocytosis checkpoints or a combinatorial MHC program — the “null case” hypothesis is false, and regeneration must be reclassified as another application of the antigenic architecture rather than a case outside it.

7. Open question: what closes the regenerative window in mammals

The P1 → P7 transition in the mouse is not a gradual quantitative shift but a switch between two qualitatively different macrophage programs within a narrow time window. This connects directly to the framework's existing interest in geroconversion and mTOR desynchronization (already documented elsewhere in the framework for aging) — possibly the same family of mechanisms that determines a cell's entry into or exit from the BGR regime also determines the switch of the macrophage program between regenerative and fibrotic. This is not developed in the present article but is recorded as a connecting question for future work: closure of the regenerative window and geroconversion may be the same process at different levels of organization (cellular and tissue).

Summary table: the four boundary cases

Case	Relation to architecture	Key evidence	Verdict
Autoimmunity (§1)	<i>Confirms by contradiction</i>	Barrier breach (orchitis, sympathetic ophthalmia); clearance failure (ANA/SLE); insufficient Treg suppression	One of three existing layers fails — no new mechanism
Transplantation (§2)	<i>Confirms by engineered borrowing</i>	Recombinant HLA-G5 reduces rejection via the same ILT-2/ILT-4 receptors used by the trophoblast	Ordinary rejection = absence of the combinatorial program; therapy recreates it artificially
Chronic infection (§3)	<i>Confirms via reversed-discovery pattern</i>	T-cell exhaustion and PD-1 first defined in chronic LCMV, later transferred to oncology	Third case of this pattern (after IDO and mCRPs in Article 5) — no longer looks accidental
Regeneration (§4–§7)	<i>Probable case outside the architecture</i>	Macrophages necessary and sufficient (axolotl); adaptive immunity dispensable (Rag1 ^{-/-}); P1 vs P7 mouse heart — same cells, opposite outcome by timing alone	Antigen-tolerance problem may not apply at all — strongest and least secure claim in the

Case	Relation to architecture	Key evidence	Verdict
			series

Synthesis

The four cases tested in this article behave differently, and that difference is informative. Autoimmunity, transplantation, and chronic infection (Part I) confirm the architecture of Articles 1–5 in three different ways — failure, borrowing, and rediscovery — without requiring an extension of the apparatus. Regeneration (Part II) inverts the very hierarchy of adaptive and innate immunity established for cancer and pregnancy, and, on preliminary analysis, may lie entirely outside the antigenic architecture rather than being another application of it. The framework hypothesis of regeneration as a “null case” is formulated with an explicit falsification criterion precisely because it is the strongest, and therefore least secure, claim in the entire Framework 1.4 series — and should be tested first, before further conclusions are built on it.

Note: the labels [Established] / Framework observation / Framework hypothesis correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4.

References

1. Recombinant HLA-G as tolerogenic immunomodulant in experimental small bowel transplantation. PLOS One. 2016. PMC4942037.
2. Biological characteristics of HLA-G and its role in solid organ transplantation. Frontiers in Immunology. 2022. doi:10.3389/fimmu.2022.902093.
3. Barber DL, Wherry EJ, Masopust D, Zhu B, Allison JP, Sharpe AH, Freeman GJ, Ahmed R. Restoring function in exhausted CD8 T cells during chronic viral infection. Nature. 2006;439(7077):682–687. doi:10.1038/nature04444.
4. Immune exhaustion: past lessons and new insights from lymphocytic choriomeningitis virus. PMC6410286.
5. Godwin JW, Pinto AR, Rosenthal NA. Macrophages are required for adult salamander limb regeneration. Proc Natl Acad Sci USA. 2013;110(23):9415–9420. doi:10.1073/pnas.1300290110.
6. Adaptive immunity is dispensable for appendage regeneration in axolotl and zebrafish (Rag1^{-/-} models). 2024.
7. Bolaños-Castro LA, et al. Immunity in salamander regeneration: where are we standing and where are we headed? Dev Dyn. 2021. PMC8363947. [Corrected on review: present in the reference list but not cited in the text; now cited in Section 4 alongside refs. 5 and 8.]
8. Scar-free wound healing and regeneration in amphibians: immunological influences on regenerative success. ScienceDirect. 2014.
9. Aurora AB, Porrello ER, Tan W, Mahmoud AI, Hill JA, Bassel-Duby R, Sadek HA, Olson EN. Macrophages are required for neonatal heart regeneration. J Clin Invest. 2014;124(3):1382–1392. doi:10.1172/JCI72181.
10. Lymphatic-macrophage crosstalk during neonatal mouse heart regeneration and transition to fibrotic repair. Eur Heart J. 2024.

11. Cardiac lymphatics retain LYVE-1-dependent macrophages during neonatal mouse heart regeneration. *Nat Cardiovasc Res*. 2025.
12. Regulatory T-cells regulate neonatal heart regeneration by potentiating cardiomyocyte proliferation in a paracrine manner. *Theranostics*. 2019.

Article 7: Active Tactics for Reducing Immune Pressure

Immunology of the Cellular Clone

Abstract

Articles 1–6 showed that the clone either inherits already-existing physiological tolerance programs (embryonic, combinatorial) or neutralizes specific surveillance mechanisms layer by layer (Article 4). This article treats a third type of tactic — neither concealment nor neutralization of the executive mechanism, but depletion of the effector resource at the level of circulating antibody. Both mechanisms discussed — local antigen overload and antigenic modulation — are established on therapeutic antibodies and on a murine leukemia model respectively; transferring either to natural antitumor surveillance remains a framework hypothesis requiring its own falsification criterion.

1. Introduction

The mechanisms treated in Article 4 (phagocytosis checkpoints, the glycocalyx, mCRPs, the metabolic layer) operate at the level of the target cell: they render the cell itself invisible or invulnerable. The tactic examined here works differently — it does not protect the target cell directly, but depletes or diverts the effector resource itself (the antibody) before it reaches the target. This is a fundamentally different level of intervention, which is why it is treated in a separate article rather than folded into the layer map of Article 4.

2. Local antigen overload (antigen sink)

[Established] The “antigen sink” phenomenon is well documented specifically on therapeutic antibodies directed against CD47.^[1]

Ubiquitous CD47 expression on healthy cells (erythrocytes, platelets) means the great majority of administered antibody binds outside the tumor: achieving serum levels of the anti-CD47 antibody magrolimab in non-human primates sufficient for xenograft efficacy required doses of 10–30 mg/kg simply to saturate this sink, before a therapeutic concentration ever reached the target — a finding that directly motivated the clinical priming/maintenance dosing strategy now used to manage the resulting anemia.^[2]

A comparable effect has been shown for soluble mesothelin (sMSLN): elevated peritoneal-fluid concentrations of mesothelin in ovarian cancer patients illustrate the scale of shed antigen that can act as a decoy, reducing the efficacy of anti-mesothelin constructs even at saturating concentrations.^[3,4]

Framework hypothesis. Transferring this mechanism from the therapeutic context (where the sink is created by widespread expression of the target on healthy tissue) to natural antitumor surveillance would require the clone itself to generate a local excess of soluble or membrane-bound antigen that depletes a pool of endogenous, rather than externally administered, antibodies or receptors. This is a logical extension, but a qualitatively different claim: the therapeutic sink is a side effect of the target's broad expression in the body, whereas the framework hypothesis requires active, directed overproduction of antigen by the clone itself as a defensive strategy.

Falsification criterion: *if tumor-clone antigen shedding is shown not to correlate with the local concentration of specific endogenous antibodies or effector cells in the microenvironment (i.e., the antigen excess produces no measurable local depletion), the hypothesis is false and the antigen sink should be treated as a therapeutic artifact rather than a natural tactic of the clone.*

3. Antibody depletion / antigenic modulation

[Established] A classic experiment showed that mouse leukemia cells of the TL phenotype, grown in the presence of cytotoxic anti-TL antiserum, lose the TL antigen from their surface — presumably through shedding or internalization of the receptor complex — becoming simultaneously insensitive to both the antibody and complement-mediated lysis; when the antiserum is removed, the antigen reappears. This phenomenon, first described for the TL antigen, was subsequently recognized as a general mechanism of antibody action, relevant to viral and autoimmune disease as well as to tumor escape and the therapeutic action of monoclonal antibodies.^[5,6]

Framework hypothesis. Generalizing to “the clone systemically depletes the pool of specific antibodies/effectors as a population-level strategy” goes beyond what the sources demonstrate: the original experiment shows local, cell-autonomous modulation of antigen under antibody pressure, not systemic depletion of the antibody pool at the level of the organism or cell population.

Falsification criterion: *if antigenic modulation under natural immune surveillance is found to be limited strictly to a cell-autonomous response by individual cells to local contact with antibody/effector (as in the original TL experiment), rather than showing signs of coordinated, population-level depletion of the resource at the tissue or organism level, the systemic version of the hypothesis should be rejected in favor of the narrow, cell-autonomous one.*

Summary table: the two tactics compared

	Local antigen overload (§2)	Antibody depletion / antigenic modulation (§3)
Established on	Therapeutic anti-CD47 antibodies (magrolimab); anti-mesothelin constructs	Murine TL-antigen leukemia model (1960s)
Mechanism	Widespread target expression on healthy tissue creates a decoy sink that absorbs administered antibody before it reaches the tumor	Antigen is shed/internalized from the cell surface under direct antibody pressure, locally and reversibly
Level of effect (as established)	Systemic — sink is body-wide (erythrocytes, platelets)	Cell-autonomous — local response of individual cells to contact with antibody
Framework hypothesis	The clone itself actively generates a local antigen excess to deplete endogenous effectors	The clone systemically depletes the antibody/effector pool as a population-level strategy
What the transfer requires	Shift from passive side-effect of broad expression → active, directed clone strategy	Shift from local, cell-autonomous modulation → systemic, population-level depletion
Falsification criterion	No correlation between clone antigen shedding and local antibody/effector depletion → sink is a therapeutic artifact, not a clone tactic	Modulation found strictly cell-autonomous, with no tissue/organism-level coordination → reject the systemic version

4. Synthesis

Both mechanisms are established firmly, but on different material: the antigen sink on therapeutic antibodies against CD47/mesothelin, antigenic modulation on a 1960s murine leukemia model. The framework hypothesis in both cases is the transfer of an established but narrow phenomenon to a broader, not directly observed context of natural antitumor surveillance, and in both cases the transfer entails a shift from a passive/local effect to an active/systemic strategy of the clone — it is this shift that needs to be tested, not the mere existence of the underlying mechanism.

Note: the labels [Established] / Framework hypothesis correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4.

References

1. Sun J, Chen Y, Lubben B, Adebayo O, Muz B, Azab AK. CD47-targeting antibodies as a novel therapeutic strategy in hematologic malignancies. *Leuk Res Rep.* 2021;16:100268. doi:10.1016/j.lrr.2021.100268.
2. Maute R, Xu J, Weissman IL. CD47–SIRP α -targeted therapeutics: status and prospects. *Immunooncol Technol.* 2022;13:100070. doi:10.1016/j.iotech.2022.100070.
3. Targeted CD47 checkpoint blockade using a mesothelin-directed antibody construct for enhanced solid tumor-specific immunotherapy. *Cancer Immunol Immunother.* 2025. (discusses the antigen-sink risk from shed mesothelin, citing ref. 4 below).
4. Okła K, Surówka J, Frąszczak K, Czerwonka A, Kaławaj K, Wawruszak A, Kotarski J, Wertel I. Assessment of the clinicopathological relevance of mesothelin level in plasma, peritoneal fluid, and tumor tissue of epithelial ovarian cancer patients. *Tumour Biol.* 2018;40(10):1010428318804937. doi:10.1177/1010428318804937.
5. Old LJ, Stockert E, Boyse EA, Kim JH. Antigenic modulation. Loss of TL antigen from cells exposed to TL antibody. Study of the phenomenon in vitro. *J Exp Med.* 1968;127(3):523–539. doi:10.1084/jem.127.3.523.
6. Chatenoud L, Bach JF. Antigenic modulation — a major mechanism of antibody action. *Immunology Today.* 1984;5(1):20–25. doi:10.1016/0167-5699(84)90061-6.

Article 8: Methods of Tumor Immunotherapy

Immunology of the Cellular Clone

Abstract

Articles 3–6 described how the natural mechanisms of immune surveillance are built and how the clone neutralizes them. This article is the mirror complement: a survey of therapeutic methods that artificially restore or substitute for those same mechanisms. The classification is organized not by the chemical nature of the agent but by mode of action — releasing a brake on an existing response versus adding a new effector the patient lacked or had in insufficient supply. Approval counts in a fast-moving field date quickly; each figure below is given with the date of the source it is drawn from rather than presented as current.

1. Introduction

Every method reviewed below can be traced to a specific node in the architecture already built: checkpoint inhibitors release precisely the brakes the clone engages in Layer 3.1 (Article 4); CAR-T and TIL exogenously supply the same granule-exocytosis/Fas-FasL mechanism described in Article 3, Section 3; oncolytic viruses and phagocytosis-checkpoint blockers strike directly at Layer 3.2. This is not a coincidence — by definition, therapy only works where a specific node is known, one that must either be unblocked or supplied.

2. Checkpoint inhibitors

[Established] Release brakes on an existing response rather than adding a new one: anti-PD-1/PD-L1, anti-CTLA-4, anti-LAG-3. As of July 31, 2023, the FDA had approved eleven such agents, each targeting one of three T-cell brakes (CTLA-4, PD-1/PD-L1, or LAG-3); at least one checkpoint inhibitor is approved for treating 20 cancer types, plus any solid tumor characterized by one of three specific molecular features, regardless of site of origin.^[1]

Framework observation: this is a direct therapeutic inversion of Layer 3.1 (Article 4) — the same PD-L1 the clone uses to switch off a CD8⁺ T cell while simultaneously braking a macrophage (Article 4, Section 3.3) becomes the drug target rather than the escape mechanism.

Note on the pendulum (cross-reference)

A subset of patients on checkpoint therapy show the opposite of the intended effect — hyperprogressive disease. This is treated as a distinct phenomenon in Article 9, Section 4.2, and is not developed further here; this article covers the intended mode of action only.

3. Adoptive cell therapy

[Established] CAR-T (chimeric antigen receptor) and TIL therapy (tumor-infiltrating lymphocytes) do not release a brake but supply the patient with specifically redirected T cells. As of a December 2024 review, the FDA had approved six CAR-T products for six hematologic malignancies (B-cell acute lymphoblastic leukemia, large B-cell lymphoma, follicular lymphoma, mantle cell lymphoma, chronic lymphocytic leukemia, and multiple myeloma); no CAR-T product was FDA-approved for a solid tumor as of that review, though two other T-

lymphocyte-based therapies had recently gained approval for solid indications (melanoma and synovial cell sarcoma) using non-CAR engineered T-cell receptors.^[2]

Framework observation: the executive mechanism here is the same granule-exocytosis and Fas/FasL pathway described in Article 3, Section 3, only delivered exogenously and typically engineered to bypass at least part of Layer 3.1 — for example, a CAR does not depend on MHC-I presentation, and is therefore resistant to its downregulation.

4. Bispecific antibodies (T-cell engagers)

[Established] Physically bring a T cell and a tumor cell together, marking the latter for destruction. The first such agent, blinatumomab, received accelerated approval in 2014; as of November 2023 a review counted seven clinically approved T-cell engager therapies on the market, and the count has continued to grow rapidly since, with additional approvals in 2024–2025 (e.g., tarlatamab for small cell lung cancer, linvoseltamab for multiple myeloma) — the exact current number is not given here since it is likely to be outdated by the time this article is read.^[3]

Framework observation: structurally this is an engineered version of ADCC (Article 3, Section 4) — except that instead of the NK cell's Fc receptor, the linking role is taken by a purpose-built bispecific protein, holding the T-cell receptor with one arm and the tumor antigen with the other.

5. Therapeutic vaccines

[Established] Expose the immune system to tumor-associated antigens, training it to recognize and attack cancer cells. Sipuleucel-T, approved by the FDA in 2010 for metastatic castration-resistant prostate cancer, remains the most established example of this class; melanoma has been the other principal proving ground for antigen-specific cancer vaccines, given its high mutational burden and well-characterized tumor antigens.

Framework observation: a direct application of the antigen categories from Article 1 — a vaccine works only where a neoantigen or oncofetal antigen (Article 4, Section 2.2) exists with sufficient immunogenicity for a targeted response to be trained against it.

6. Cytokine therapy

[Established] High-dose IL-2 and IFN- α were among the earliest FDA-approved cancer immunotherapies (for metastatic melanoma and renal cell carcinoma), stimulating the proliferation and activity of immune cells directly, independent of recognition of any specific antigen; IL-12 and IL-15 analogues have since been explored on the same principle.

Framework observation: this is the only category in this list not tied to a specific node of the antigenic architecture — rather than releasing a brake or delivering an effector, it crudely amplifies the system as a whole, including Th1 polarization (Article 3, Section 5), which accounts for both its broad effect and its systemic toxicity.

7. Monoclonal antibodies

[Established] Rituximab (anti-CD20, approved 1997) and trastuzumab (anti-HER2, approved 1998) were among the first monoclonal antibodies approved for cancer and work mainly through ADCC rather than checkpoint blockade.

Framework observation: the same logic as Section 4, but without a bispecific intermediary — the antibody directly opsonizes the target, engaging the natural ADCC mechanism of the effector cell (Article 3, Section 4).

8. Oncolytic viruses and phagocytosis-checkpoint blockers

[Established] Two newer categories: oncolytic viruses and developmental blockers of “don't eat me” signals (CD47–SIRPα, CD24–Siglec-10). Talimogene laherparepvec (T-VEC), approved by the FDA in 2015 for advanced melanoma, remains the only FDA-approved oncolytic virus therapy in the United States.^[4]

Framework observation: phagocytosis-checkpoint blockers are a direct therapeutic application of Layer 3.2 (Article 4), already treated in detail there.

Framework hypothesis (labeled as inference, not established for oncolytic viruses specifically): oncolytic viruses cause immunogenic cell death, which by the general mechanism described in Article 1, Section 6 would be expected to expose nuclear material and potentially trigger a secondary adaptive response against tumor neoantigens following viral lysis. This is a plausible extension of an already-established mechanism, not a claim directly demonstrated for oncolytic virotherapy in the sources consulted for this article, and should be verified against the oncolytic-virus-specific literature before being treated as established.

Summary table: all eight methods

Method	Mode of action	Node (Articles 1/3/4)	Key fact
Checkpoint inhibitors (§2)	<i>Release a brake</i>	Layer 3.1 — PD-1/PD-L1, CTLA-4, LAG-3	11 agents approved as of Jul 2023; ≥20 cancer types
Adoptive cell therapy: CAR-T/TIL (§3)	<i>Deliver an effector</i>	Art. 3 §3 — granule exocytosis/Fas-FasL, exogenous	6 CAR-T products, 6 malignancies, as of Dec 2024
Bispecific antibodies (§4)	<i>Deliver an effector</i>	Art. 3 §4 — engineered ADCC bridge	First approval 2014 (blinatumomab); 7 agents as of Nov 2023, growing
Therapeutic vaccines (§5)	<i>Deliver an effector (trains it)</i>	Art. 1 — requires an immunogenic neoantigen/oncofetal antigen	Sipuleucel-T, 2010, prostate cancer
Cytokine therapy (§6)	<i>Amplify the system (no specific node)</i>	None — the one exception in this list	IL-2/IFN-α among the earliest approved (melanoma, RCC)
Monoclonal antibodies (§7)	<i>Deliver an effector</i>	Art. 3 §4 — natural ADCC, no bispecific bridge	Rituximab (1997), trastuzumab (1998)
Oncolytic viruses (§8)	<i>Deliver an effector (lysis + secondary response)</i>	Art. 1 §6 — immunogenic cell death (inference, not established for OVs)	T-VEC (2015) — only FDA-approved OV therapy
Phagocytosis-	<i>Release a brake</i>	Layer 3.2 — CD47–SIRPα, CD24–	Developmental / not yet approved

Method	Mode of action	Node (Articles 1/3/4)	Key fact
checkpoint blockers (§8)		Siglec-10	

9. Synthesis

The eight categories reduce to the two modes of action stated in the introduction (Section 8 explicitly comprises two categories — oncolytic viruses and phagocytosis-checkpoint blockers — rather than one), but the specific point of application for each traces back to a node already described in Articles 1, 3, and 4 — meaning this article introduces no new material but shows that the therapeutic arsenal against cancer is, in essence, a mirror image of the architecture already assembled, not a toolkit independent of it. The one exception is cytokine therapy, which is not tied to a specific node, which also accounts for its cruder effect and toxicity profile relative to the other seven.

Note: the labels [Established] / Framework observation / Framework hypothesis correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4.

References

1. AACR Cancer Progress Report 2023. Immunotherapy: Pushing the Frontier of Cancer Medicine. American Association for Cancer Research.
2. Brudno JN, Maus MV, Hinrichs CS. CAR T Cells and T-Cell Therapies for Cancer: A Translational Science Review. JAMA. 2024;332(22):1924–1935. doi:10.1001/jama.2024.19462.
3. T-cell engagers — the structure and functional principle and application in hematological malignancies. Int J Mol Sci / MDPI. 2024. PMC11048836. (seven approved T-cell engager therapies as of November 2023).
4. FDA Approves First Oncolytic Virus Therapy (talimogene laherparepvec/T-VEC/Imlygic) for Melanoma. National Cancer Institute news release, 2015. [Government news source, not a peer-reviewed article — cited for the fact of approval only.]

Article 9: Causes of Low Efficacy and Paradoxical Effects of Cancer Immunotherapy

Immunology of the Cellular Clone

Abstract

Three structurally distinct mechanisms explain treatment failure and paradoxical effects in cancer therapy. Dormancy (Article 4, Section 3.2) is a reserve that escapes detection. Classical resistance (Article 4, Section 5.2) is adaptation of the clone under pressure. The third, treated in depth here, is activation of the regeneration pendulum: a single executive program (macrophage M2 polarization, TGF- β , PGE2/VEGF-mediated angiogenesis) that normally restores damaged tissue can be engaged by therapy through two proposed triggers — cell death and direct immune redistribution upon checkpoint release. The evidence base for the two triggers is uneven and is flagged as such throughout: the cell-death trigger is well established, but so far only in cytotoxic and radiotherapy contexts, not immunotherapy specifically; the immune-redistribution trigger rests on a single documented case plus a review-level hypothesis. Both gaps are treated as open questions rather than settled facts.

1. Introduction

The standard classification of treatment failure — primary resistance, acquired resistance, an undetectable reserve — describes why therapy fails to control the clone. This article adds a case in which therapy does not merely fail but actively provokes progression, through the same physiological circuit described in Article 6 for regeneration. It also flags, rather than glosses over, two evidentiary gaps in that argument: the extrapolation from cytotoxic/radiotherapy data to immunotherapy specifically, and the thinness of direct support for the immune-redistribution trigger.

2. Dormant cells (brief)

[Established] Already treated in Article 4, Section 3.2 — the equilibrium phase of immunoediting maintains immunogenically “silent” clone variants, undetectable by therapy aimed at actively proliferating or highly immunogenic cells. This is a reserve that exists independently of treatment, not a response to it.

3. Classical resistance (brief)

[Established] Already treated in Article 4, Section 5.2 — primary resistance (antigen loss, HLA loss, constitutive PD-L1), adaptive resistance (T-cell dysfunction, co-expression of inhibitory checkpoints), acquired resistance (relapse after an initial response). This is adaptation of the clone under the pressure of ongoing therapy, not a side effect of the act of treatment itself.

Summary table: three causes of treatment failure

Cause	Nature	Evidentiary status
Dormancy (§2)	A reserve of immunogenically silent clone	<i>Established (Article 4, §3.2)</i>

Cause	Nature	Evidentiary status
	variants existing independently of treatment	
Classical resistance (§3)	Adaptation of the clone under the pressure of ongoing therapy	<i>Established (Article 4, §5.2)</i>
Regeneration pendulum (§4)	Activation of a physiological tissue-repair program by the act of intervention itself, not by evasion or adaptation	<i>Least secure of the three — cell-death trigger established only outside immunotherapy; immune-redistribution trigger rests on one case</i>

4. Activation of the regeneration pendulum: one executive node, two proposed triggers

4.1. Trigger A — cell death (the Phoenix Rising pathway)

[Established] Apoptotic cells were first shown to activate a caspase-3–dependent signaling cascade — named the “phoenix rising” pathway — to promote compensatory proliferation as a normal wound-healing and tissue-regeneration mechanism.^[1]

Tumor cells dying under cytotoxic or radiation therapy do not simply disappear: active caspase-3 in apoptotic cells activates calcium-independent phospholipase A2 (iPLA2), releasing prostaglandin E2 (PGE2), which stimulates proliferation of surviving tumor cells via STAT3 signaling. Higher levels of activated caspase-3 in human tumor tissue correlate with increased rates of recurrence and death, and the same regenerative pathway that heals normal tissue is usurped by the tumor.^[2]

Separately, caspase-3 in tumor cells dying after irradiation was shown to drive post-irradiation angiogenesis (PIA) via VEGF-A, likely through Akt signaling; inhibiting caspase-3 suppressed both the proangiogenic effect in vitro and in vivo and tumor growth in a xenograft model — meaning the act of cell death itself, rather than survival of a resistant clone, restores the tumor's blood supply after therapy.^[3]

Scope note (added on review): all three sources above establish this pathway for radiotherapy and cytotoxic chemotherapy, not for immunotherapy specifically. None of them tests whether immune-mediated cell death — via perforin/granzyme or Fas/FasL (Article 3, Section 3) — triggers the same caspase-3 → PGE2/VEGF-A cascade as DNA-damage-induced apoptosis from irradiation. Caspase-3 is a shared downstream executioner across many apoptotic routes, so the extrapolation to immunotherapy is plausible, but it is an extrapolation, not a demonstrated fact, and is labeled as a framework hypothesis rather than folded silently into the established mechanism above.

[Framework hypothesis] immune-mediated tumor cell death (CTL/NK-driven, Article 3 Section 3) activates the same caspase-3 → iPLA2 → PGE2/VEGF-A cascade as radiation- or chemotherapy-induced apoptotic death, and therefore contributes to tumor repopulation and revascularization following immunotherapy by the same mechanism.

Falsification criterion: *if tumors undergoing predominantly immune-mediated killing (e.g., during CAR-T or checkpoint-inhibitor therapy) show caspase-3 activation without the downstream PGE2/VEGF-A signature seen after irradiation, or show no correlation between caspase-3 activity and repopulation/revascularization in this context, the extrapolation is false*

and the phoenix rising pathway should be treated as specific to genotoxic/cytotoxic cell death rather than a general property of caspase-3-mediated apoptosis regardless of trigger.

4.2. Trigger B — immune redistribution upon checkpoint release

[Established] 4 to 29% of patients on anti-PD-1/PD-L1 therapy show hyperprogressive disease (HPD) — a paradoxical acceleration of tumor growth — with a comparative cohort study finding HPD notably more frequent with checkpoint inhibitors than with chemotherapy in pretreated NSCLC, pointing to a mechanism specific to release of this particular brake rather than a general property of anticancer therapy.^[4]

The existence and clinical significance of HPD itself rests on solid, cohort-level evidence (ref. 4) and is treated as established. The specific molecular mechanism proposed below does not carry the same weight and is labeled accordingly.

Proposed, weakly supported: candidate mechanisms include expansion of PD-1+ regulatory T cells and a shift of tumor-associated macrophages toward an M2-like phenotype; in one documented fatal case of HPD in metastatic bladder cancer, omics analysis of autopsy tissue implicated M2 macrophages, TGF- β signaling, and interleukin-8 in the pathogenesis. This mechanism rests on a single case report plus a review-level hypothesis, not on a cohort or mechanistic study, and should not be read as carrying the same evidentiary weight as Trigger A.^[5,6]

4.3. One pendulum, or two separate mechanisms?

Framework observation: the executive node proposed for both triggers is the same in kind. M2 macrophage polarization and TGF- β already appeared in Article 6, Section 5 as markers of the fibrotic-regenerative branch of the macrophage response; IL-8 is a pro-angiogenic chemokine whose downstream effect on vascularization is broadly comparable to that of the VEGF-A described in Section 4.1, though the two act through distinct receptor systems (CXCR1/2 versus VEGFR) and IL-8 additionally recruits neutrophils — the parallel is functional, not molecular identity.

The difference proposed between the two triggers is that Trigger A requires cell death as the primary event (caspase-3 is activated inside the dying cell itself), while Trigger B is thought to be triggered by direct immune redistribution (Treg expansion → macrophage shift) rather than by cell death.

Gap noted on review: no source cited in this article directly establishes that HPD onset precedes substantial tumor cell death. The clinical literature documents early acceleration of tumor growth by imaging at the first on-treatment assessment (typically 6–8 weeks), not the absence of significant cell death up to that point. The claim that Trigger B operates “before substantial cell death” is therefore a plausible reading of the clinical timeline, not a demonstrated fact, and the falsification criterion below has been reworded to make the missing measurement explicit rather than assumed.

[Framework hypothesis] hyperprogression is not a separate, independent mechanism but a second route of activation of the same regeneration pendulum as Trigger A, triggered by immune redistribution instead of cell death.

Falsification criterion distinguishing the two hypotheses: *this requires directly measuring caspase-3/PGE2 activity in tumor tissue from patients with HPD, at a time point matched against the degree of tumor cell death present. If elevated caspase-3/PGE2 activity is found even in early HPD with minimal measurable cell death, this favors a single pathway with two*

triggers converging upstream of the executive node. If the M2 shift and HPD phenotype are found with demonstrably normal caspase-3 activity, these are two structurally distinct pathways converging only at the executor (M2/TGF- β /angiogenesis). Neither measurement has yet been reported in the literature surveyed for this article, so the question remains genuinely open rather than provisionally resolved in either direction.

Summary table: the two proposed triggers compared

	Trigger A — cell death	Trigger B — immune redistribution
Established for	Cytotoxic chemotherapy and radiotherapy (not immunotherapy specifically)	Existence/incidence of HPD itself (cohort-level, ref. 4)
Proposed executor	Caspase-3 → iPLA2 → PGE2 / VEGF-A (Phoenix Rising)	PD-1+ Treg expansion → M2 macrophage shift → TGF- β /IL-8
Primary event	Cell death (caspase-3 activated inside the dying cell)	Immune redistribution, proposed to precede substantial cell death (timing not directly measured)
Evidentiary weight	Well established, but extrapolated to immunotherapy as framework hypothesis	Weak — one case report + review-level hypothesis; not cohort or mechanistic data
Falsification criterion	Immune-mediated killing shows caspase-3 activation without PGE2/VEGF-A signature → extrapolation false	Direct caspase-3/PGE2 measurement in HPD tissue needed — result decides single-pathway vs. two-pathway model (§4.3)

5. Synthesis

The three causes of treatment failure differ in logical kind: dormancy is a reserve existing independently of treatment; classical resistance is adaptation of the clone under the pressure of ongoing therapy; the regeneration pendulum is activation of a physiological tissue-repair program triggered by the act of intervention itself, not by evasion or adaptation on the clone's part. This third mechanism is the least secure of the three as currently evidenced: its cell-death trigger is well established for cytotoxic and radiation therapy but only hypothesized, not shown, for immunotherapy; its immune-redistribution trigger rests on a single case rather than a cohort. The practical implication — that part of treatment failure should be sought in whether the attack itself switches on a recovery program, not only in whether the clone survived it — is worth pursuing, but the two experiments named in Sections 4.1 and 4.3 need to be run before this mechanism can be elevated from framework hypothesis to established fact.

Note: the labels [Established] / Framework observation / Framework hypothesis correspond to the claim-marking system adopted for the whole framework starting with Framework 1.4.

References

1. Li F, Huang Q, Chen J, Peng Y, Roop DR, Bedford JS, Li CY. Apoptotic cells activate the “phoenix rising” pathway to promote wound healing and tissue regeneration. *Sci Signal.* 2010;3(110):ra13. doi:10.1126/scisignal.2000634.
2. Huang Q, Li F, Liu X, Li W, Shi W, Liu FF, O'Sullivan B, He Z, Peng Y, Tan AC, Zhou L, Shen J, Han G, Wang XJ, Thorburn J, Thorburn A, Jimeno A, Raben D, Bedford JS, Li CY. Caspase 3-

mediated stimulation of tumor cell repopulation during cancer radiotherapy. *Nat Med.* 2011;17(7):860–866. doi:10.1038/nm.2385.

3. Feng X, Tian L, Zhang Z, Yu Y, Cheng J, Gong Y, Li CY, Huang Q. Caspase 3 in dying tumor cells mediates post-irradiation angiogenesis. *Oncotarget.* 2015;6(32):32353–32367. doi:10.18632/oncotarget.5898.
4. Ferrara R, Mezquita L, Texier M, Lahmar J, Audigier-Valette C, Tessonnier L, Mazieres J, Zalcman G, Brosseau S, Le Moulec S, Leroy L, Duchemann B, Lefebvre C, Veillon R, Westeel V, Koscielny S, Champiat S, Ferte C, Planchard D, Remon J, Boucher ME, Gazzah A, Adam J, Bria E, Tortora G, Soria JC, Besse B, Caramella C. Hyperprogressive disease in patients with advanced non-small cell lung cancer treated with PD-1/PD-L1 inhibitors or with single-agent chemotherapy. *JAMA Oncol.* 2018;4(11):1543–1552. doi:10.1001/jamaoncol.2018.3676.
5. How can immune checkpoint inhibitors cause hyperprogression in solid tumors? *Frontiers in Immunology.* 2020. PMC7098964. [Review-level hypothesis, not a mechanistic or cohort study — see Section 4.2.]
6. Mechanistic insights into lethal hyperprogressive disease induced by a PD-L1 inhibitor in metastatic urothelial carcinoma. PMC11408499. [Single case report — see Section 4.2.]

Afterword

This monograph marks the completion of one stage of a long intellectual journey rather than its conclusion. Whether the ideas presented here ultimately prove correct will be decided not by the author, but by future experimental work and scientific discussion.

I would like to express my sincere gratitude to my faithful AI research assistants, Claude (Anthropic) and ChatGPT. Throughout this project, they served not as replacements for scientific thinking, but as tireless partners in literature exploration, critical analysis, and the organization of complex interdisciplinary ideas.

My deepest thanks go to my daughter, Sarah-Lana Demi, Data Scientist, for her patience, understanding, and unwavering support throughout this work, and for enduring my often difficult character during the many months devoted to this project. I am especially grateful to her for her help in studying and understanding stochastic processes, which proved essential to the formal treatment developed in this monograph.

I am also deeply grateful to my colleagues at Hillel Yaffe Medical Center, Israel, for their encouragement, professional advice, and for generously sharing their knowledge and clinical experience:

Professor Ilan Bruchim

Dr. Haim David

Dr. Ariel Polonsky

Finally, I wish to thank everyone who will read this work with an open and critical mind.

Science advances not only through answers, but through questions that are worth asking.